

# Covid Data Analysis

## Data Science

Sakshi Patel  
Ravya Vangaveti  
Trisha Chandrashekar  
Sai Surya Teja Reddy Bommepalli

### I. STAGE I - DATA AND PROJECT UNDERSTANDING

This project aims to analyze COVID-19 cases and deaths at the county level in the United States. By integrating datasets related to county populations, COVID-19 confirmed cases, and deaths, the analysis attempts to uncover insights into how the virus spread and affected different counties. The dataset merges these three key variables and serves as a foundation for further analysis.

#### A. Datasets Used

##### 1. COVID-19 County Population Dataset

Provides population data for U.S. counties, essential for calculating case and death rates per 100,000 people.

##### 2. COVID-19 Confirmed Cases Dataset

Tracks daily confirmed cases of COVID-19 at the county level.

##### 3. COVID-19 Deaths Dataset

Tracks daily COVID-19-related deaths at the county level.

#### B. Key Variables

##### COVID-19 County Population Dataset

- **FIPS Code (County Identifier):** Unique numeric identifier for each county. (Integer or String)
- **County Name:** Name of the county. (String)
- **State Name/Code:** Name or two-letter code of the state. (String)
- **Population:** Total population of the county. (Integer)

##### COVID-19 Confirmed Cases Dataset

- **FIPS Code (County Identifier):** Numeric code for each county. (Integer or String)
- **County Name:** Name of the county. (String)
- **State Name/Code:** Name or two-letter code of the state. (String)
- **Date:** Date when the cases were recorded. (Date or String; Format: YYYY-MM-DD)
- **Confirmed Cases:** Number of confirmed COVID-19 cases. (Integer)

##### COVID-19 Deaths Dataset

- **FIPS Code (County Identifier):** Numeric code for each county. (Integer or String)
- **County Name:** Name of the county. (String)
- **State Name/Code:** Name or two-letter code of the state. (String)

- **Date:** Date when deaths were recorded. (Date or String; Format: YYYY-MM-DD)
- **Deaths:** Number of COVID-19-related deaths. (Integer)

#### C. Steps in COVID-19 Data Integration

- 1) Reading and Displaying COVID-19 Datasets
- 2) Understanding Key Variables and Data Types
- 3) Merging the Datasets
- 4) Final Super COVID-19 Dataframe

#### D. Final Super COVID-19 Dataframe Columns

- County FIPS Code
- County Name
- State
- Population
- Daily Confirmed Cases
- Daily Deaths

#### E. Enrichment Datasets for COVID-19

- **Census Demographic ACS:** Sakshi Patel
- **ACS Social, Economic, and Housing:** Sai Surya
- **Employment Dataset:** Ravya Vangaveti
- **Presidential Election Results:** Trisha Chandrashekar

##### Census Demographic ACS

About Dataset:

This American Community Survey (ACS) produces population, demographic and housing unit estimates for 2020. The 2020 Census provides the official counts of the population and housing units for the counties Executive Summary:

This report documents the process of cleaning, merging, and analyzing COVID-19 and demographic data from U.S. counties. The main focus is on the COVID-19 pandemic trends for 2020, particularly in Arizona, as well as enriching the COVID-19 data with demographic information from the American Community Survey (ACS) to better understand the spread of the virus. The steps outlined include data preparation, cleaning, and merging to create a comprehensive dataset. The enriched dataset helps explore various social, economic, and demographic factors that may have influenced the virus's transmission. This analysis allows us to pose key hypotheses regarding the relationship between demographic variables and the spread of COVID-19. Work:

1. Data Cleaning
2. Merging the COVID-19 Data
3. Analyzing COVID-19 Data for 2020
4. COVID-19 Trends for the Last Week of 2020 (Arizona)

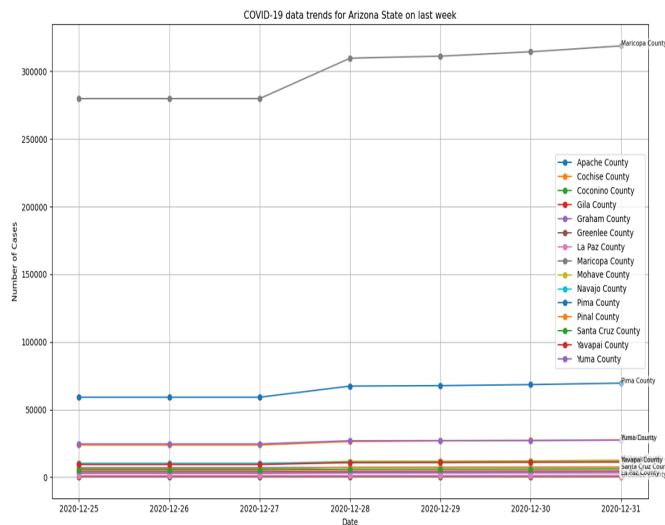


Fig. 1. COVID-19 Trends for the Last Week of 2020 (Arizona)

## 5. Enrichment Data: ACS Demographic and Housing Estimates

### 6. Cleaning ACS Data

Exclusion of Puerto Rico Data

Remove Prefix from 'Geography' Column

Rename the 'Geography' Column to 'countyFIPS' Column

Change Datatype of 'countyFIPS' column

Drop the last column which contains NaN value

### 7. Merging the Enriched Dataset

Final Merged Dataset: 3144 rows and 1052 columns

### 8. Enrichment Data's Role in COVID-19 Spread Analysis

### 9. Initial Hypothesis Questions

[a] Does higher population density correlate with a higher rate of COVID-19 cases?

[b] Are counties with a larger elderly population experiencing higher COVID-19 death rates?

[c] Does the sex ratio influence the COVID-19 death rate?

## ACS Social, Economic, and Housing

### About Dataset:

This is American Community Survey (ACS) produces population, demographic and housing unit estimates for 2020. The 2020 Census provides the official counts of the population and housing units for the counties.

### Executive Summary:

This report documents the process of cleaning, merging, and analyzing COVID-19 and demographic data from U.S. counties. The main focus is on the COVID-19 pandemic trends for 2020, particularly in Arizona, as well as enriching the COVID-19 data with demographic information from the American Community Survey (ACS) to better understand the spread of the virus. The steps outlined include data

preparation, cleaning, and merging to create a comprehensive dataset. The enriched dataset helps explore various social, economic, and demographic factors that may have influenced the virus's transmission. This analysis allows us to pose key hypotheses regarding the relationship between demographic variables and the spread of COVID-19.

The dataset you linked to is from the American Community Survey (ACS) and is specifically the ACS Demographic and Housing Estimates (DP05). This dataset provides detailed social, economic, housing, and demographic information for various geographic areas in the United States<sup>2</sup>. Here are some key points about this dataset:

**Geographic Coverage:** The data covers a wide range of geographic areas, including states, counties, metropolitan/micropolitan statistical areas, places, ZIP Code Tabulation Areas, and neighborhoods (census tracts).

**Data Profiles:** The DP05 table is part of the ACS Data Profiles, which summarize the most frequently requested social, economic, housing, and demographic data for a single geographic area. It provides both estimates and percentages.

**Release Frequency:** The ACS data is released annually, and the DP05 table is part of the 1-Year Estimates.

**Data Use:** This dataset is critical for government programs, policies, and decision-making, as it provides comprehensive statistics about the nation.

### What It Includes:

- **Population Estimates:** Total population counts along with estimates by age, sex, race, and Hispanic or Latino origin.
- **Household Estimates:** Data on the number of households and average household size.
- **Race and Ethnicity:** Detailed breakdowns of race categories and Hispanic or Latino origin.
- **Age Distribution:** Population estimates by different age ranges, such as under 18, 18-64, and 65 and over.

### How to Access and Use the Data:

- **Search and Filter:** You can search for specific data by geographic area, such as states, counties, and cities.
- **Download Options:** Data is available for download in multiple formats, including CSV and Excel, making it easy to analyze in your preferred software.
- **Visualization Tools:** Interactive tools and maps are available to help visualize the data, making it easier to understand trends and patterns.

### Steps:

#### • Data Cleaning

This involves removing or correcting any inaccuracies in the raw data. It may include:

- Identifying and dealing with missing data.
- Removing duplicates.
- Correcting inconsistent data entries.
- Ensuring that all data follows the same format.

#### • Merging the COVID-19 Data

Combining multiple COVID-19 datasets into a single dataset. This could involve:

- Aligning data from different sources.
- Matching data based on common identifiers (e.g., county FIPS codes).
- Ensuring all data is in the same format.

#### • Analyzing COVID-19 Data for 2020

Performing an in-depth analysis of the cleaned and merged COVID-19 data. This may include:

- Calculating infection, recovery, and death rates.
- Identifying patterns and trends over time.
- Comparing data across different counties or states.

#### • COVID-19 Trends for the Last Week of 2020 (North Carolina)

Focusing on the trends in Arizona for the final week of 2020. This could involve:

- Plotting infection rates, death rates, and other relevant metrics.
- Analyzing changes in these metrics over this specific period.

#### • Enrichment Data: ACS Demographic and Housing Estimates

Incorporating demographic and housing data from the ACS to enrich the COVID-19 data. This includes:

- Retrieving relevant demographic data (e.g., population density, age distribution).

#### • Cleaning ACS Data

Specific cleaning steps to prepare ACS data for merging:

- Exclusion of Puerto Rico Data: Removing data entries for Puerto Rico.
- Remove Prefix from 'Geography' Column: Removing any prefixes to ensure consistent naming.
- Rename 'Geography' Column to 'countyFIPS' Column: Renaming the column to a standard identifier.
- Change Datatype of 'countyFIPS' Column: Ensuring the data type is consistent (e.g., converting to numeric).
- Drop the Last Column with NaN Values: Removing the final column if it contains unnecessary NaN values.

#### • Merging the Enriched Dataset

Combining the cleaned COVID-19 data with the cleaned ACS data. Resulting in a comprehensive dataset with:

- 3,144 rows (representing counties).
- 1,052 columns (including COVID-19 and demographic data).

#### • Enrichment Data's Role in COVID-19 Spread Analysis

Using the enriched dataset to explore how demographic factors influence COVID-19 spread. This could involve:

- Formulating and testing hypotheses.
- Identifying significant correlations between demographic variables and COVID-19 metrics.

- Understanding the socio-economic factors that impact the virus's transmission and outcomes.

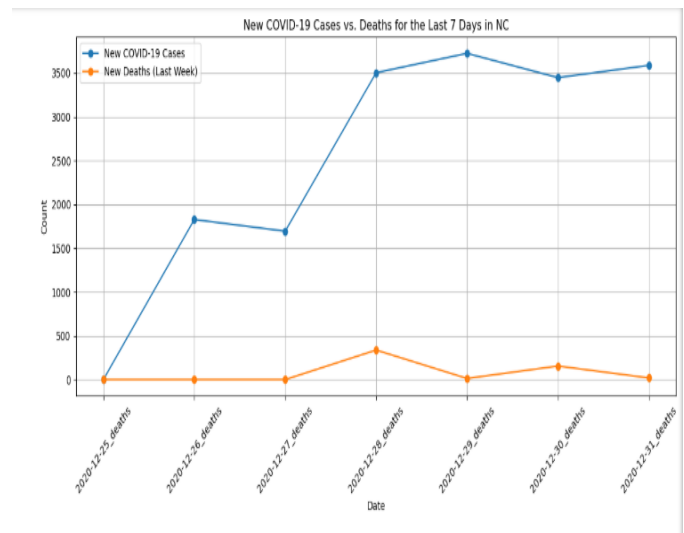


Fig. 2. COVID-19 Trends for the Last Week of 2020 (North Carolina)

#### Initial Hypothesis Questions:

- Do areas with higher average household income experience lower COVID-19 infection rates?
- Is there a significant difference in COVID-19 recovery rates between urban and rural counties?
- Does the level of educational attainment in a county affect the COVID-19 vaccination rates?

#### Employment Dataset

**Enrichment Data set- Employment dataset** Employment data set :

It has three data sets they are :

- County-ests-employment-and-wages
- State-ests-employment-and-wages
- Top-ten-ests-employment-and-wages

#### Steps followed :

Step 1:

- Understanding the above data set
- Reading the data set
- Printing first five rows of dataset

Step 2:

- Printing the variable dictionary to understand the variables present in each column.

Step 3:

- Filtering the data with the year 2020.

Step 4:

- Merging Covid-19 data set with our Enrichment dataset which is filtered for the year 2020.

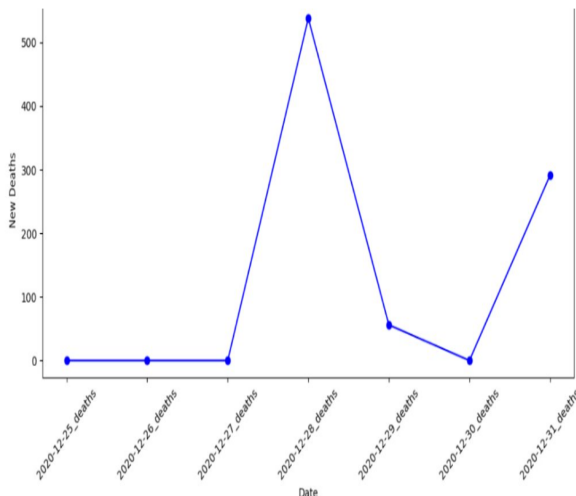


Fig. 3. Enrichment Dataset

Step 5:

- Saving the final merge dataset as a .csv file

Step 6:

- Initial Hypothesis Questions

Step 7:

- Selecting the state
- Filtering the data for NY

Step 8:

- Identifying the last week's trend for the specified state that we have selected.

**Initial Hypothesis:**

- How did the employment sector get effected due to Covid19?
- Did the employment rates is affected in essential sectors like health care, regular services during the pandemic?
- Did you think the areas with higher employment faced more COVID-19 cases than the areas with lower employment?
- If major cities faced more number of covid cases due to employment, how explain?
- Does employment rate effect the economy of the county?

## Presidential Election Results

About Dataset:

This dataset contains detailed US presidential election outcomes for 2020. It includes information on the states and counties in the USA, along with the candidates competing in each area, their respective parties, the total votes received, and whether the candidate won in the particular state or county. Executive Summary:

This report outlines the process of cleaning, merging, and analyzing both the COVID-19 data and the US presidential election dataset. The primary focus is on analyzing COVID-19 trends in 2020 and comparing them with 2023,

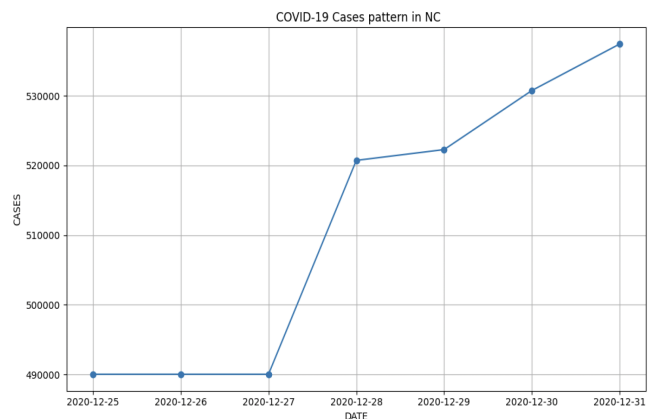
particularly in North Carolina. Additionally, the report explores how enriching the COVID-19 dataset with election data might reveal insights into any potential connection between the virus spread and election variables. The steps include data preparation, cleaning, and merging to create a comprehensive dataset, which enables us to form key hypotheses about the relationship between the election and COVID-19 spread. Work:

1. **Data Cleaning**

2. **Merging the COVID-19 Data**

3. **Analyzing COVID-19 Data for 2020**

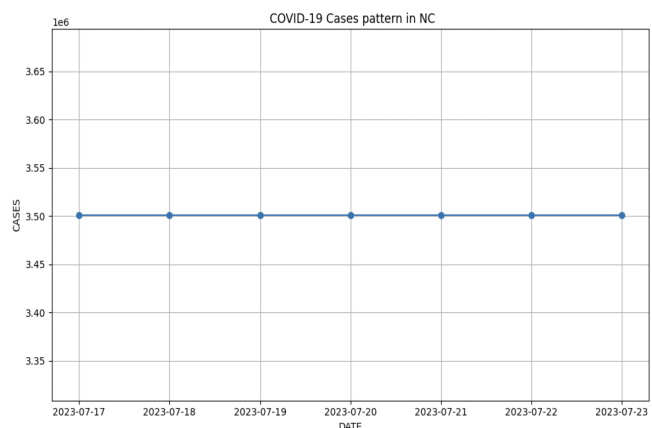
4. **COVID-19 Cases Trends for the Last Week of 2020 (North Carolina)**



We can clearly see the spike in the number of cases closer to the new year (last week of 2020), So we can also propose a hypothesis that the increase in cases trend could be because of the new year celebration gathering and people traveling

Fig. 4. COVID-19 CasesTrends for the Last Week of 2020 (North Carolina)

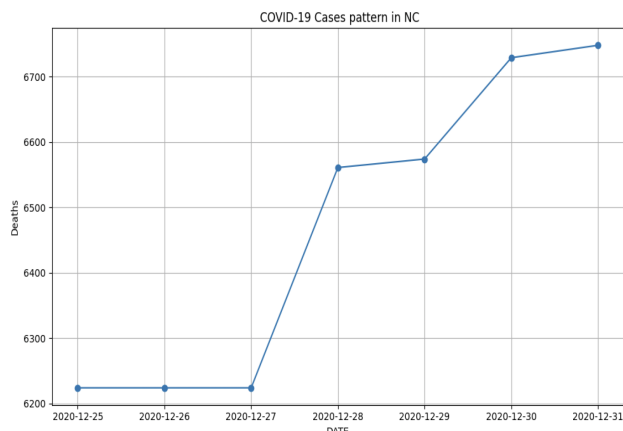
5. **COVID-19 Cases Trends for the Last Week of 2023 (North Carolina)**



From the Above graph its clearly visible the the number of cases of covid-19 has been constant throughout the last week of the year 2023 to compare lets check for the year 2

Fig. 5. COVID-19 Cases Trends for the Last Week of 2023 (North Carolina)

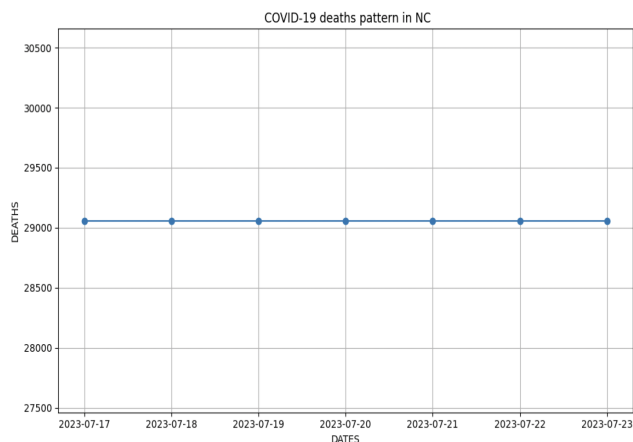
## 6. COVID-19 Deaths Trends for the Last Week of 2020 (North Carolina)



We can clearly see the spike in the number of deaths closer to the new year (last week of 2020)

Fig. 6. COVID-19 Deaths Trends for the Last Week of 2020 (North Carolina)

## 7. COVID-19 Deaths Trends for the Last Week of 2023 (North Carolina)



From the Above graph its clearly visible the the number of deaths of covid-19 has been constant throughout the last week

Fig. 7. COVID-19 Deaths Trends for the Last Week of 2023 (North Carolina)

## 8. Enrichment Data: Presidential Election Results

### 9. Merging the Enriched Dataset

Merged Dataset based on State and County

## 10. Understanding Enrichment Data's Role in COVID-19 Spread Analysis

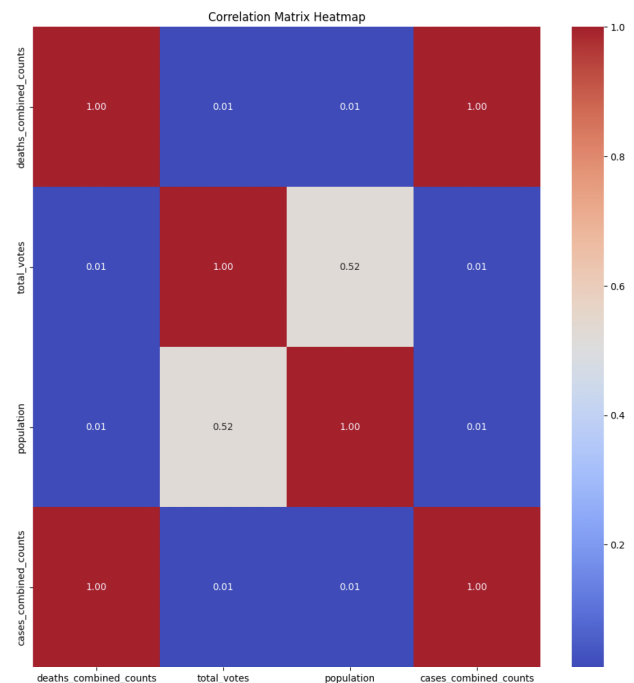
- The election dataset offers insights into how political affiliations (such as Republican and Democratic) might have influenced COVID-19 outcomes.
- Case and Death Rates: For example, counties with a higher percentage of Republican or Democratic votes

may have experienced varying rates of COVID-19 cases or deaths

- Does the level of educational attainment in a county affect the COVID-19 vaccination rates?

## 11. Initial Hypothesis Questions

- We can explore whether there is any correlation between political affiliation and the spread of COVID-19 cases or deaths. For example, we could investigate whether counties that supported the Republican candidate had higher COVID-19 case rates, or if counties favoring the Democratic candidate experienced lower COVID-19 death rates, which may provide insights into adherence to COVID-19 regulations by different political groups.
- We can examine if there's any relationship between the victory of a particular candidate and the spread of COVID-19 cases or deaths.
- Additionally, we can analyze whether there is any correlation between the total votes cast and COVID-19 case rates or deaths.



We see a relation between cases and death count but will have to perform more in depth analysis to see the correlation proposed in our hypothesis

Fig. 8. Correlation Heatmap between my enrichment dataset and covid-19 dataset)

## II. STAGE II - DATA MODELING

1. Compare the weekly statistics (mean, median, mode) for the number of cases and deaths across the US. Filtering the data from 2020.6.1(Monday) - 2021.1.3(Sunday).

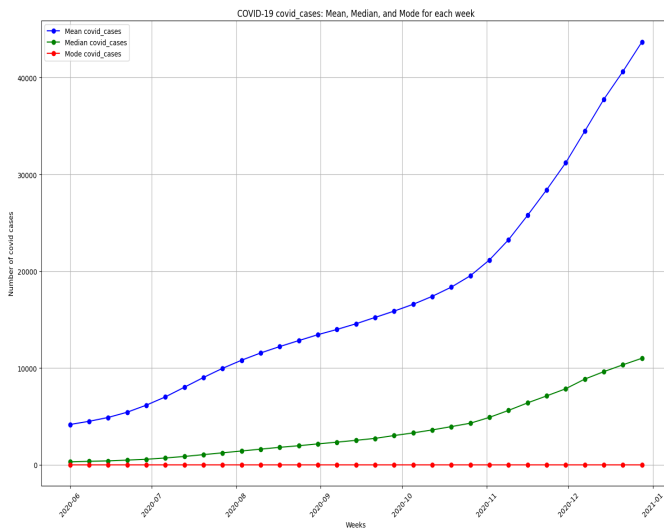


Fig. 9. COVID-19 Cases: Mean, Median, and Mode for each week

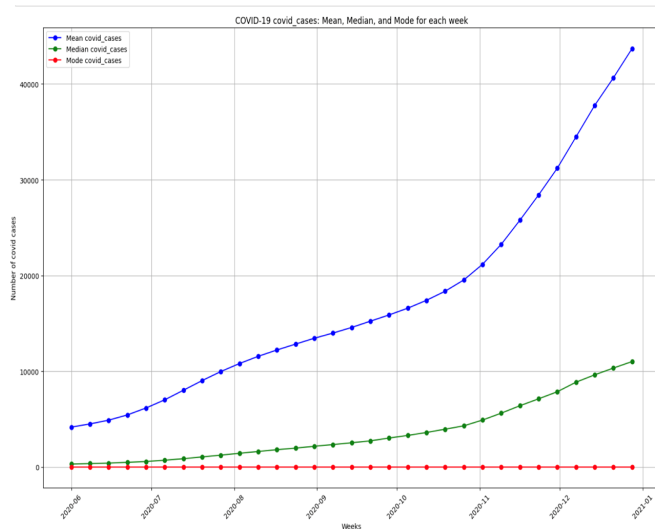


Fig. 10. COVID-19 Deaths: Mean, Median, and Mode for each week

2. Compare the data against other countries of the world.

We selected below eight countries:

South America  
Indonesia  
Pakistan  
Nigeria  
Brazil  
United States  
India  
Canada

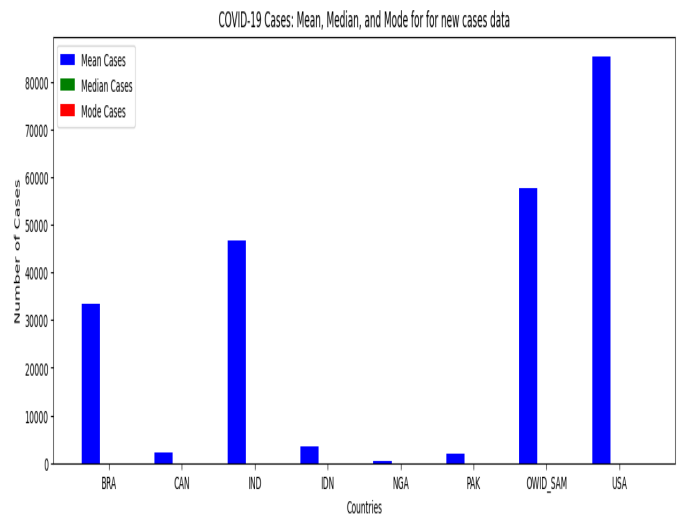


Fig. 11. COVID-19 Cases: Mean, Median, and Mode for new cases data

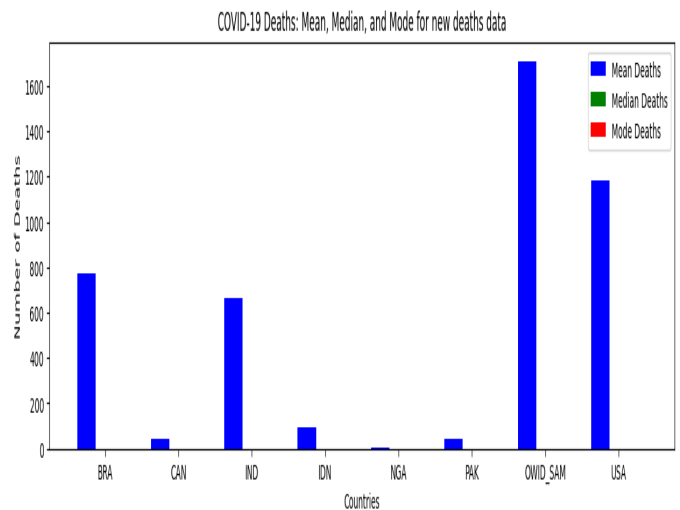


Fig. 12. COVID-19 Deaths: Mean, Median, and Mode for new deaths data

3. Plot weekly trends (cases and deaths) of countries

In this task, we need to plot weekly trends by utilizing aggregate, normalized by population, difference in cases, and log normalized values.

4. Background research

Try to do background research to see what the reasoning for the peaks can be:

**United States** - Peak Cases: Around July 5, 2020, a sharp increase in cases. Because the 4th is a holiday, when gatherings and travel might happen so the graph has increased.

**India** - Peak Cases by: June 28, increased because at this time they could have fewer restrictions, or it might align with local holidays or an increase in domestic travel.

**Indonesia** Peak cases on July 5, 2020, increased this is because of Eid celebrations where there would be a lot of gatherings.

**Nigeria** June 21, 2020, This because due to the gradual decrease in strictness of lockdown restrictions, or it could correlate with religious festivals or increased travel.

**Pakistan** Peak Cases increased cases in June 21, 2020 this is because of Eid celebrations, which often involve large gatherings and inter-city travel.

| COVID Cases                                                                                                                                                                                                                                                                                                                                                                                                     | COVID Deaths                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Differences in Case Rates Across States</p> <p><b>Arizona:</b> Arizona shows the sharpest increase in cases, especially starting around November 2020.</p> <p><b>California (CA), Florida (FL), and Georgia (GA):</b> These states show similar trends but have lower peaks compared to Arizona.</p> <p><b>New York (NY):</b> New York has a significantly lower case rate compared to the other states.</p> | <p>Differences in Death Rates Across States</p> <p><b>New York (NY):</b> New York has the highest number of deaths per 100,000.</p> <p><b>Arizona (AZ):</b> Arizona shows a significant increase in death rates starting around October 2020.</p> <p><b>California (CA), Florida (FL), and Georgia (GA):</b> These states follow a similar upward trend, with California and Florida showing the next highest death rates after Arizona.</p> |
| <p><b>Case Peaks:</b> The peaks in Arizona, California, Florida, and Georgia around November to January align with the national trend of a surge in COVID-19 cases during the fall and winter of 2020 due to increased social interactions and holiday gatherings.</p>                                                                                                                                          | <p><b>Death Peaks:</b> New York's higher death rate may reflect its earlier experiences during the first wave of the pandemic in 2020. Arizona's rapid rise in both cases and deaths.</p>                                                                                                                                                                                                                                                    |

Fig. 13. Comparison Across 5 Different States

#### 4. County-Level Analysis in Arizona:

Weekly trends for these counties were plotted, both as raw values and using log normalization to identify patterns of exponential growth.

#### Census Demographic ACS

Work:

##### 1. Data Overview:

The dataset used for this analysis is 'super\_covid19.csv', containing 3142 rows and 2535 columns. The data covers all U.S. counties and provides daily statistics for COVID-19 cases and deaths. The analysis focused on data from June 1, 2020 (Monday) to January 3, 2021 (Sunday). This period was selected to observe the COVID-19 cases and deaths.

##### 2. Weekly Statistics (Arizona State):

Cases:

- Mean: The mean, or average number of cases, shows a continuous rise over time.
- Median: The median, which represents the middle value of the dataset, also increases but at a much slower rate compared to the mean.
- Mode: The mode, or the most frequent number of cases reported in a week, remains constant and very low throughout the entire period.

Deaths:

- Mean: The mean continues to rise throughout the graph, with a sharp upward trend starting around November 2020.
- Median: The median also rises but at a much slower rate than the mean.
- Mode: The mode remains quite low, indicating that most weeks or regions had consistently low death counts.

##### 3. Comparison Across State:

Five states (Arizona, California, Florida, Georgia, and New York) were compared in terms of weekly cases and deaths. The data was normalized by population (per 100,000 people) to ensure comparability across states with varying populations.

Top 5 counties with high cases:

|     | County Name     | total_cases |
|-----|-----------------|-------------|
| 103 | Maricopa County | 30690035    |
| 106 | Pima County     | 5552644     |
| 110 | Yuma County     | 2789651     |
| 107 | Pinal County    | 2365829     |
| 105 | Navajo County   | 1305635     |

Fig. 14. Top 5 counties with high cases

Top 5 counties with high deaths:

|     | County Name     | total_deaths |
|-----|-----------------|--------------|
| 103 | Maricopa County | 611918       |
| 106 | Pima County     | 121212       |
| 110 | Yuma County     | 62978        |
| 105 | Navajo County   | 46031        |
| 104 | Mohave County   | 42181        |

Fig. 15. Top 5 counties with high deaths

Description of COVID Cases raw values and log normalized values:

- What is causing the trends? The population of Maricopa County, Arizona's most populous county, likely contributed to its higher case numbers. Pima, Yuma, Pinal, and Navajo counties also show growth, at smaller scales. These counties are less populated, resulting in fewer cases.
- Peaks: The clear peak in Maricopa County's raw cases occurs around late December 2020 to early January 2021. This could be due to holiday travel, increased gatherings, and colder weather.

- **State Pattern:** The trends in these top five counties likely follow the overall state pattern seen in Arizona. Like much of the U.S., Arizona experienced a surge in COVID-19 cases during the summer of 2020 and another significant peak during the winter months of 2020–2021.

Description of COVID-19 deaths raw values and log normalized values:

- **What is causing the trends?:** The rise in deaths corresponds with the winter wave of COVID-19 in late 2020, likely driven by increased transmission during holiday gatherings and colder weather.

- **Peaks:** The major peak for Maricopa County was around the end of 2020, while the other counties show smaller, less distinct peaks.

- **State Pattern:** The overall trends align with the state's COVID-19 pattern, where Maricopa County, being the most populous, shows the most significant impact.

## ACS Social, Economic, and Housing

**Introduction** This study focuses on the top five counties with the highest COVID-19 cases and deaths. The goal is to understand the spread and impact of the virus across these regions and to identify patterns and peaks in the data.

### Data Analysis

#### A. Top 5 Counties with the Highest Cases

- **Cook County:** Leading with 38,139,227 total cases, highlighting significant community spread and public health challenges in densely populated areas.
- **DuPage County:** Second with 5,075,064 cases, reflecting substantial virus spread in suburban regions.
- **Lake County:** 4,507,058 cases, indicating ongoing COVID-19 impact and the need for public health measures.
- **Will County:** 4,147,178 cases, emphasizing the widespread nature of the virus.
- **Kane County:** 3,688,970 cases, signaling the necessity for sustained health efforts.

#### B. Top 5 Counties with the Highest Deaths

- **Cook County:** Highest deaths at 1,186,696, underscoring the severe pandemic impact and the importance of healthcare responses.
- **DuPage County:** 132,100 deaths, showing significant pandemic consequences in the region.
- **Lake County:** 111,044 deaths, highlighting serious ramifications and the need for vigilant healthcare resources.
- **Will County:** 92,682 deaths, indicating the critical need for community health initiatives.
- **Kane County:** 76,483 deaths, marking the pandemic's toll and the importance of ongoing health efforts.

**Graph Analysis** The following graphs represent the weekly trends for cases and deaths in the top five counties, both in raw values and log-normalized values.

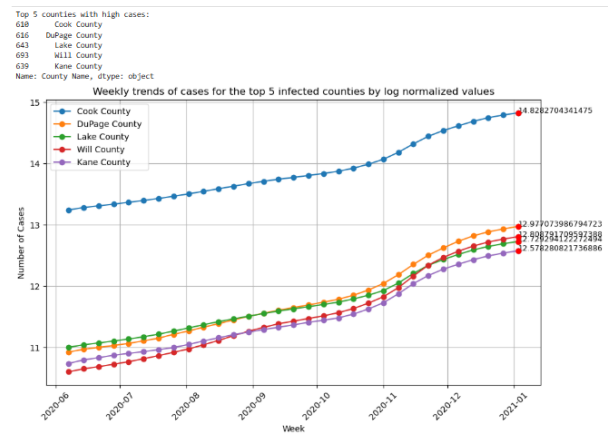


Fig. 16. Weekly COVID-19 Cases in Top Counties

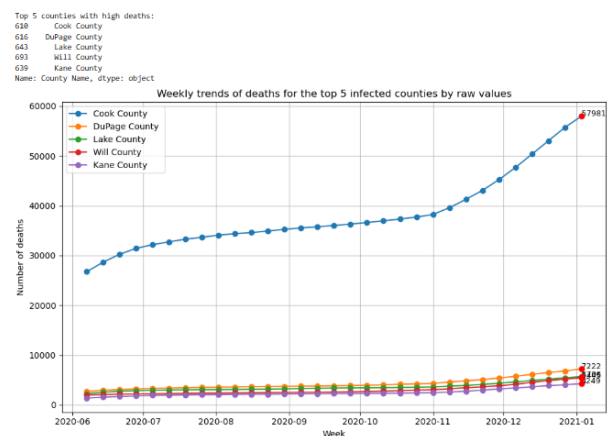


Fig. 17. Weekly COVID-19 Deaths in Top Counties

#### C. Weekly Trends for Cases and Deaths

### Discussion

The differences in COVID-19 rates across the counties can be attributed to several factors, including population density, public health measures, socio-economic conditions, and healthcare infrastructure. The peaks identified in the graphs are compared with the US national pattern to understand regional variations.

### Conclusion

This analysis provides insights into the impact of COVID-19 on the top five counties with the highest cases and deaths. It highlights the need for tailored public health strategies to mitigate the spread and reduce mortality rates.

## Employment Dataset

### Prerequisites:

- Reading the Super covid data.
- Filtering the required state data (NY state).
- Filtering NY state data from dates 2020.6.1(Monday)-2021.1.3(Sunday)

## Task 1

- Printing the values Mean Median mode for the state NY data.
- Dividing the columns as deaths and cases.
- Making two distinct frames for cases and deaths.
- Removing\_cases and\_deaths.

|            | Cases Mean    | Cases Median | Cases Mode | Deaths Mean | Deaths Median | Deaths Mode |
|------------|---------------|--------------|------------|-------------|---------------|-------------|
| 2020-06-07 | 42348.225886  | 1591.5       | 1          | 3353.258865 | 98.5          | 7           |
| 2020-06-14 | 43016.080645  | 1649.0       | 1          | 3409.564516 | 101.5         | 6           |
| 2020-06-21 | 43567.500000  | 1684.5       | 2          | 3452.193548 | 101.5         | 6           |
| 2020-06-28 | 44086.080645  | 1734.0       | 1          | 3486.741935 | 101.5         | 6           |
| 2020-07-05 | 44597.951613  | 1820.5       | 1          | 3566.080645 | 101.5         | 6           |
| 2020-07-12 | 45108.661290  | 1845.0       | 1          | 3594.177419 | 101.5         | 6           |
| 2020-07-19 | 45691.645161  | 1875.5       | 1          | 3611.693548 | 101.5         | 6           |
| 2020-07-26 | 46249.177419  | 1902.0       | 1          | 3626.177419 | 101.5         | 7           |
| 2020-08-02 | 46778.451613  | 1964.5       | 1          | 3637.548387 | 101.5         | 8           |
| 2020-08-09 | 47299.338710  | 2031.5       | 1          | 3646.354839 | 101.5         | 7           |
| 2020-08-16 | 47807.387097  | 2081.0       | 1          | 3653.580645 | 101.5         | 7           |
| 2020-08-23 | 48304.935484  | 2117.0       | 1          | 3660.983871 | 101.5         | 5           |
| 2020-08-30 | 48784.822581  | 2155.5       | 1          | 3667.790323 | 101.5         | 5           |
| 2020-09-06 | 49352.177419  | 2185.0       | 2          | 3674.048387 | 101.5         | 5           |
| 2020-09-13 | 49910.645161  | 2314.0       | 1          | 3679.677419 | 101.5         | 5           |
| 2020-09-20 | 50504.516129  | 2417.5       | 1          | 3685.338710 | 101.5         | 5           |
| 2020-09-27 | 51138.387097  | 2706.5       | 2          | 3690.080645 | 101.5         | 5           |
| 2020-10-04 | 51973.548387  | 3166.5       | 1          | 3697.403226 | 101.5         | 5           |
| 2020-10-11 | 53063.225886  | 3397.0       | 1          | 3706.983871 | 101.5         | 5           |
| 2020-10-18 | 54155.516129  | 3623.0       | 2          | 3715.854839 | 101.5         | 5           |
| 2020-10-25 | 55346.225886  | 3927.0       | 2          | 3725.741935 | 110.0         | 4           |
| 2020-11-01 | 56804.241935  | 4421.5       | 1          | 3737.500000 | 126.0         | 4           |
| 2020-11-08 | 58650.790323  | 4814.0       | 1          | 3750.451613 | 143.0         | 5           |
| 2020-11-15 | 61660.741935  | 5636.5       | 1          | 3771.290323 | 177.5         | 6           |
| 2020-11-22 | 65448.532258  | 6195.0       | 1          | 3796.467742 | 157.0         | 4           |
| 2020-11-29 | 70105.806452  | 7201.5       | 1          | 3829.580645 | 203.0         | 2           |
| 2020-12-06 | 76272.451613  | 8628.5       | 1          | 3874.467742 | 201.0         | 2           |
| 2020-12-13 | 83965.516129  | 10455.5      | 1          | 3934.758865 | 204.0         | 2           |
| 2020-12-20 | 92146.419355  | 12778.0      | 1          | 4018.806452 | 220.0         | 2           |
| 2020-12-27 | 100551.112903 | 15631.0      | 1          | 4093.854839 | 267.5         | 2           |
| 2021-01-03 | 110029.306452 | 18323.0      | 1          | 4221.790323 | 326.0         | 2           |

Fig. 18. Mean, Median, Mode for Cases and Deaths

- Transposing so that dates will be in rows not in columns.
- Adding up daily to weekly data.
- Calculating weekly statics of deaths and cases.
- Printing weekly statistics of deaths and cases.

## Task 2

- Selecting 5 states as NY they are Alaska, Colorado, Florida, New Jersey, Texas.

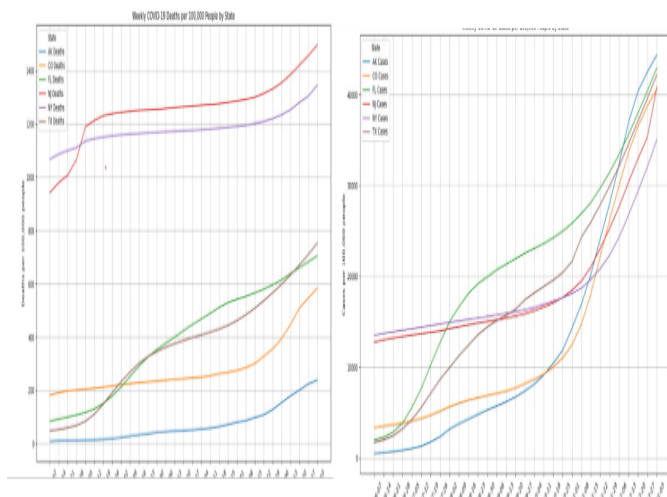


Fig. 19. Weekly Deaths by State

- Printing state and its population.

- Printing State names and dates 2020.6.1(Monday)-2021.1.3(Sunday)
- Merging the above steps on the state.
- Plotting the graph which has weeks on X - the axis and normalized cases on Y – the axis similarly for deaths.

## Task 3

- Filtering the counties in NY state with the highest rates of deaths and cases
- Filtering the date range 2020.6.1(Monday) - 2021.1.3(Sunday)
- Grouping county's name by sum of all deaths and cases in date columns.

### Top Counties with High Cases and Death Rates:

| CountyName      | Total_Cases | Total_Deaths |
|-----------------|-------------|--------------|
| Kings County    | 569536380   | 12964721     |
| Queens County   | 540901598   | 12256740     |
| Suffolk County  | 367457749   | 4295613      |
| Nassau County   | 349947363   | 3956631      |
| New York County | 345437048   | 5567748      |
| Bronx County    | 330601029   | 8034084      |

Fig. 20. Top Countries with High Cases and Deaths

- Calculating total deaths and cases of each county.
- Arranging them in order and displaying the data.

## Task 4

- Plots the graph for the above-selected counties by raw values and log normalized values.
- Graph as dates on X-axis and total deaths on Y—the axis similarly for cases.

### For Cases:

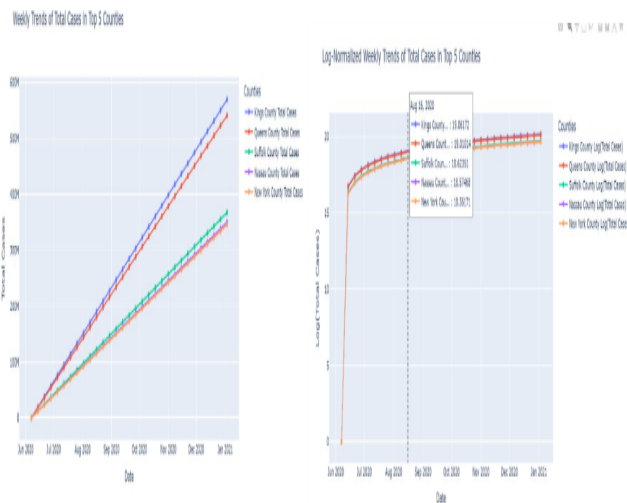


Fig. 21. For Cases

### For Deaths:

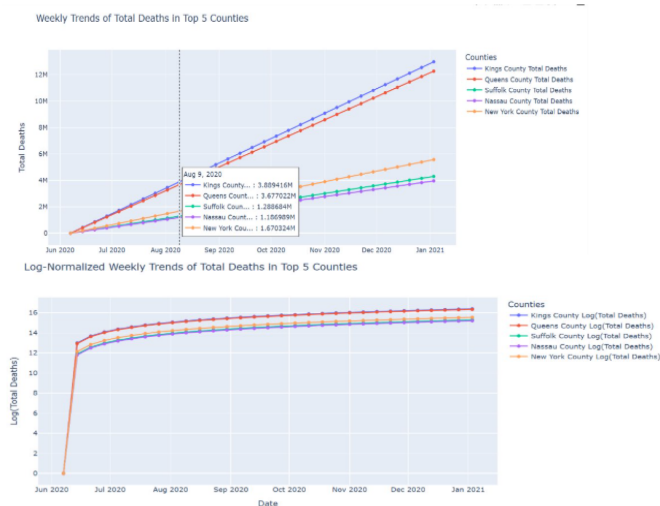


Fig. 22. Deaths

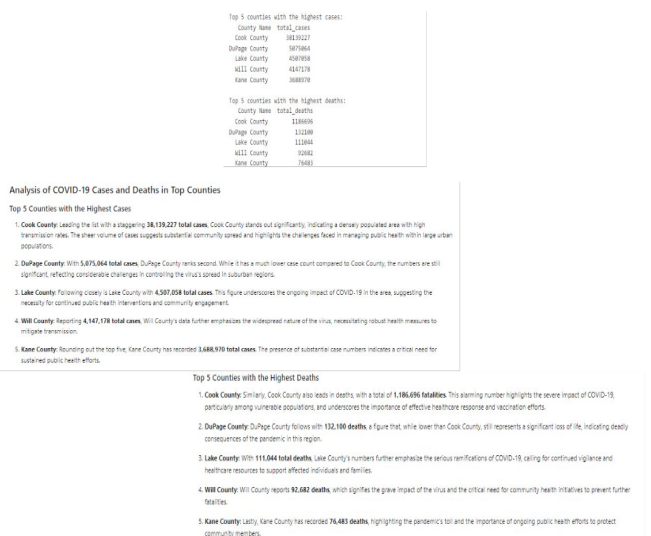


Fig. 23.

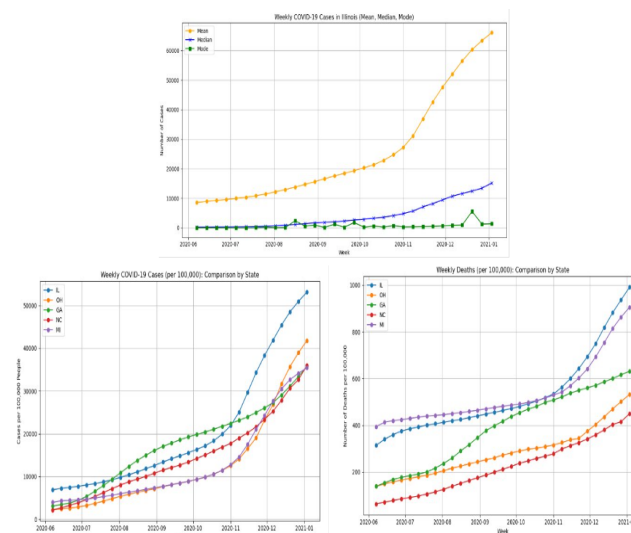


Fig. 24. Cases in illinois

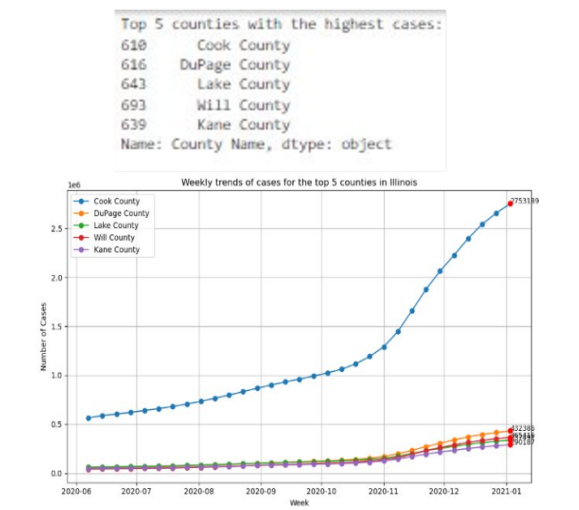


Fig. 25. Top 5 Countries

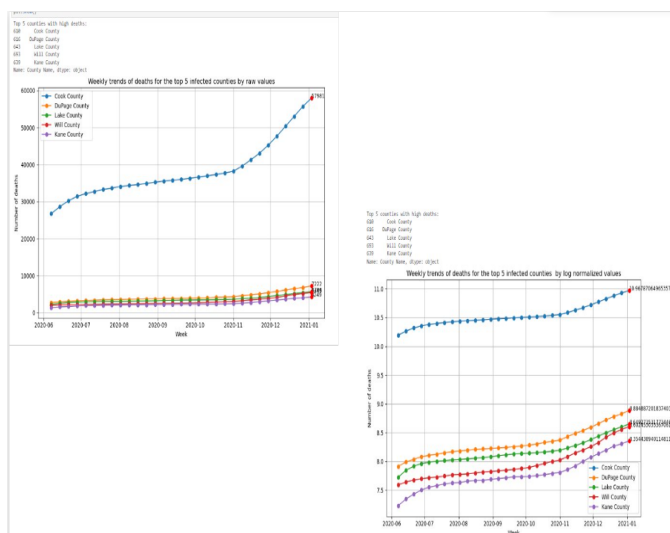


Fig. 26. Top 5 Countries of Weekly

## Presidential Election Results

Work:

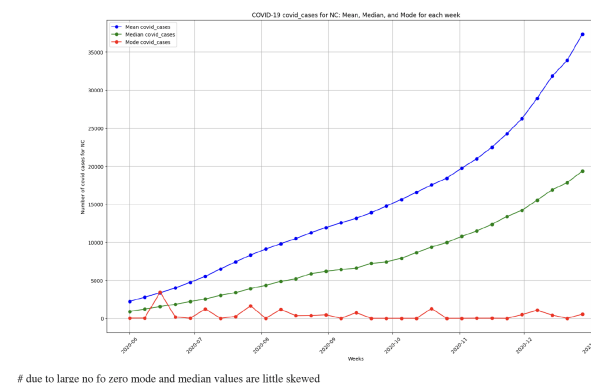
### 1. Data Overview:

The dataset used for this analysis is 'super\_covid19.csv', containing 3142 rows and 2535 columns. The data covers all U.S. counties and provides daily statistics for COVID-19 cases and deaths. The analysis focused on data from June 1, 2020 (Monday) to January 3, 2021 (Sunday). This period was selected to observe the COVID-19 cases and deaths.

### 2. Weekly Statistics (North Carolina State):

Cases:

- Mean: The average number of COVID-19 cases rises steadily, peaking around the New Year due to holiday gatherings. The highest case count was observed in December 2020.
- Median: The median, representing the middle value of the dataset, also increases but at a slower rate compared to the mean.
- Mode: The mode, or the most frequently reported number of cases in a week, stays consistently low throughout the period. A high number of zero values skews both the mode and median, slightly distorting the analysis.

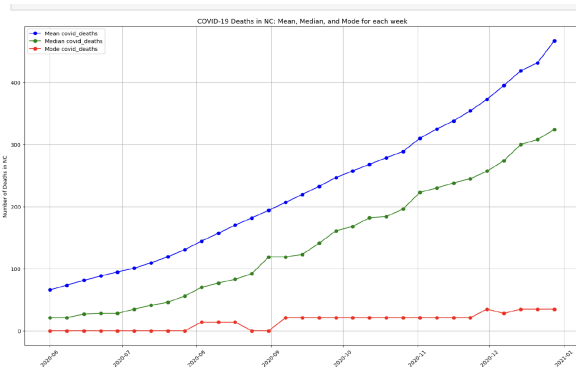


we see that there is a increase in mean cases and it peaks and the end of the year around new year ,which was the cases due to holidays and new year.we saw that covid was highest in decemeber of 2020

Fig. 27. Mean, Median, Mode for North Carolina State

Deaths:

- Mean: The average number of COVID-19 deaths shows a steady increase, with a sharp rise around the New Year, likely due to holiday gatherings. The peak death count occurred in December 2020.
- Median: The median also increases but at a slower pace compared to the mean.
- Mode: The mode remains consistently low. A significant number of zero values skew both the mode and median, causing some distortion in the analysis.



# due to large no of zero mode and median values are little skewed

we see that there is a increase in mean deaths and it peaks and the end of the year around new year ,which was the cases due to holidays and new year

Fig. 28. Mean, Median, Mode for North Carolina State

### 3. Comparison Across State:

Five states (Washington DC, Illinois, New Hampshire, North Carolina, Utah, and West Virginia ) were compared in terms of weekly cases and deaths. The data was normalized by population (per 100,000 people) to ensure comparability across states with varying populations. In both cases and deaths, we initially see low numbers, but by the end of the period, there is a noticeable increase across all states.

In particular, the death cases in Washington, D.C. appear to be consistently high, likely due to its role as a hub for people.

In West Virginia, there is a sharp rise in both deaths and cases. The number of cases seems to have never stopped increasing and continued to rise until the end of the month.

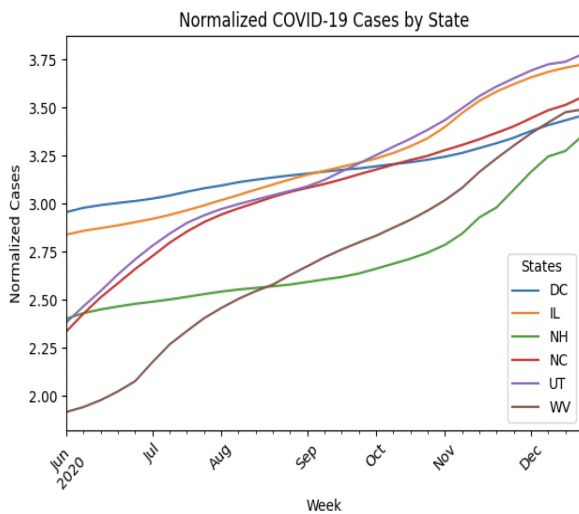


Fig. 29. Comparison of cases Across 5 Different States

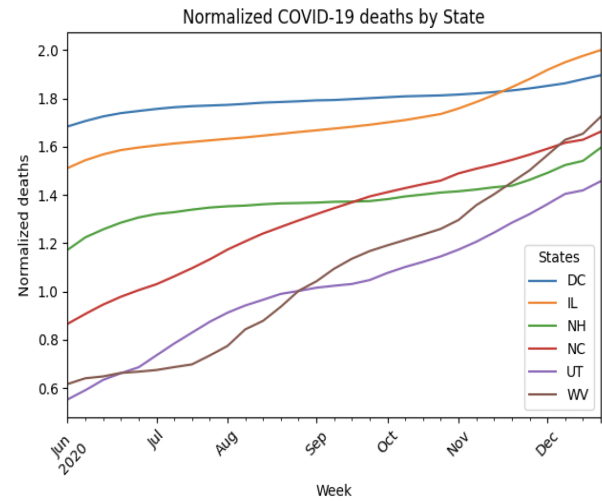


Fig. 30. Comparison of Deaths Across 5 Different States

### 4. County-Level Analysis in North Carolina:

Weekly trends for these counties were plotted, both as raw values and using log normalization to identify patterns of exponential growth.

|      | County Name        | total_cases |
|------|--------------------|-------------|
| 1983 | Mecklenburg County | 231597057   |
| 2015 | Wake County        | 231476009   |
| 1964 | Guilford County    | 98527134    |
| 1957 | Forsyth County     | 76945961    |
| 1949 | Cumberland County  | 69232513    |

Fig. 31. Top 5 counties with high cases

|      | County Name        | total_cases |
|------|--------------------|-------------|
| 1983 | Mecklenburg County | 1399599     |
| 1964 | Guilford County    | 1023233     |
| 2015 | Wake County        | 975284      |
| 1959 | Gaston County      | 683938      |
| 1957 | Forsyth County     | 664615      |

Fig. 32. Top 5 counties with high deaths

### 3. Description of COVID Cases raw values and log normalized values:

The plots for both cases and deaths reveal notable differences between normalized and raw data. This variation may be attributed to the normalization process, which transforms the data by minimizing the influence of noise and outliers, thereby emphasizing relative differences. In contrast, the raw data contains considerable noise, including actual values, outliers, and high and low values without a proper distribution. Despite these differences, both raw and normalized case/death data show an upward trend, peaking

toward the end of the year. From our analysis in Stage 2 of the project, we observe that both county and state data follow a similar pattern, with increases leading up to December and a peak at the end of that month. Overall, the COVID-19 cases and deaths across the entire country, including states and counties, consistently rose from June to December, reaching their peak at the end of December.

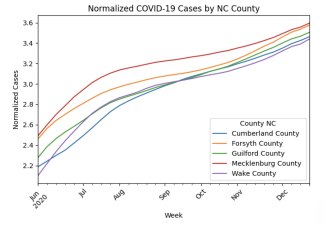


Fig. 33. Top 5 counties with high cases Normalised Data

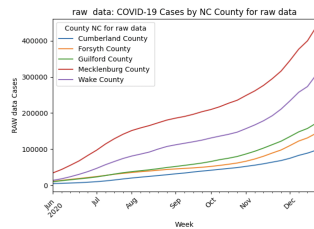


Fig. 34. Top 5 counties with high cases Raw Data

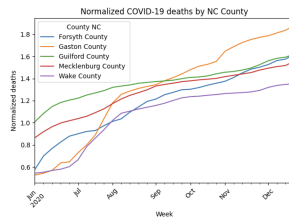


Fig. 35. Top 5 counties with high deaths Raw Data

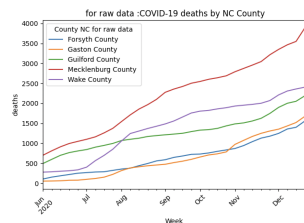


Fig. 36. Top 5 counties with high deaths Normalised Data

### III. STAGE III - DISTRIBUTIONS AND HYPOTHESIS TESTING

The project focuses on the analysis of COVID-19 cases using statistical modeling and data correlation techniques. Key objectives include fitting distributions to COVID-19

cases, modeling Poisson distributions for case counts and deaths, and exploring correlations between demographic (enrichment) variables and COVID-19 outcomes. The data used for this analysis was provided in earlier phases of the project, covering different states in the U.S.

#### Census Demographic ACS

#### 1. Fitting a Distribution to COVID-19 Case Data:

##### Data Overview:

The analysis started by extracting COVID-19 case data for the state of Arizona from a larger dataset. Data was filtered from June 1, 2020, to January 3, 2021. A total of 217 days of data was analyzed, representing daily COVID-19 cases across multiple counties within the state.

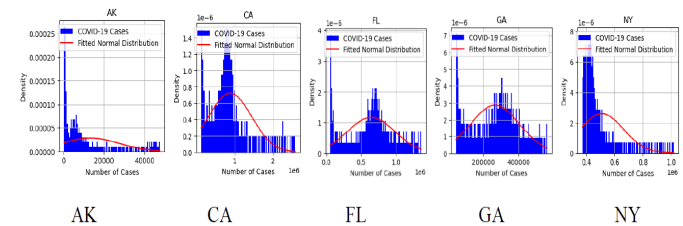


Fig. 37. Comparison with Other States

##### Comparison with Other States:

- CA and NY have more extreme data distributions with higher skewness and kurtosis, meaning they have more outliers and greater variability.
- FL and GA have more symmetric, less peaked distributions, suggesting a more even spread across their data.
- AK has the smallest variability and skewness, indicating more uniform data with less outlier presence.
- States like California and Florida experienced large and fluctuating COVID-19 outbreaks, with periods of extreme case surges, as reflected in their high means, variability, and positive skewness.
- Arizona had a moderate outbreak, with some periods of increased cases, but it was more stable compared to highly impacted states like CA.
- Alaska had a much smaller outbreak, with more consistent and lower case numbers, indicated by its low mean and low variability.
- States like New York and Georgia also experienced notable outbreaks, with NY showing more extreme events, while GA had a more stable case count.

#### 2. Modeling Poisson Distribution for COVID-19 Cases and Deaths:

##### Poisson Distribution for Case Counts:

Poisson distributions were modeled for daily new COVID-19 cases per 100,000 people in Arizona and compared to five other states. Arizona, California, and Florida displayed wider, more variable distributions, indicating fluctuating case numbers. States like Alaska and Georgia had narrower distributions, implying more stable case patterns.

##### Poisson Distribution for Deaths:

A similar Poisson distribution was modeled for COVID-19 deaths, showing that states like Alaska experienced fewer fluctuations in daily death counts, while Florida and Arizona had wider, more variable distributions.

### 3. Correlation Between Enrichment Data and COVID-19 Outcomes:

#### Hypotheses:

##### 1. Age Groups and COVID-19 Outcomes:

- Hypothesis: Counties with a larger proportion of children (5-9 years old and under 18 years) will have fewer COVID-19 deaths per 100k population.
- Reasoning: Younger populations are less likely to experience severe symptoms, which may contribute to lower mortality rates.

##### 2. Under 5 Years Population and COVID-19 Outcomes:

- Hypothesis: Counties with a higher percentage of the population under 5 years old will have fewer cases and deaths per 100k population.
- Reasoning: Infants and young children are generally at lower risk of severe COVID-19 outcomes compared to older adults, though they may contribute to transmission.

##### 3. Female Population and COVID-19 Outcomes:

- Hypothesis: Counties with a higher proportion of females will have similar or slightly lower COVID-19 death rates compared to males.
- Reasoning: Globally, men have shown a slightly higher COVID-19 death rate, though female populations may exhibit different outcomes based on cultural and socioeconomic factors.

### ACS Social, Economic, and Housing

**Distribution Fitting (Non-Poisson):** Select a state (e.g., North Carolina) and fit a distribution (e.g., Normal, Lognormal) to the number of COVID-19 cases.

- Describe the chosen distribution and why it was selected.
- Graphically plot the distribution.
- Calculate and plot the Probability Mass Function (PMF) if using discrete values.
- Describe the distribution statistics: center (mean/median), variance, skewness, and kurtosis.

**Comparison with Other States** Select five other states (e.g., California, Florida, Texas, New York, Illinois).

- Fit the same type of distribution to the COVID-19 cases in these states.
- Compare the distribution statistics across these states.
- Discuss any differences and what they imply about the spread of COVID-19 in these states.

**Poisson Distribution Modeling** Model the COVID-19 cases and deaths for a state using a Poisson distribution.

- Compare the Poisson distribution to the first distribution you fitted.
- Plot the Poisson distribution and describe its characteristics.

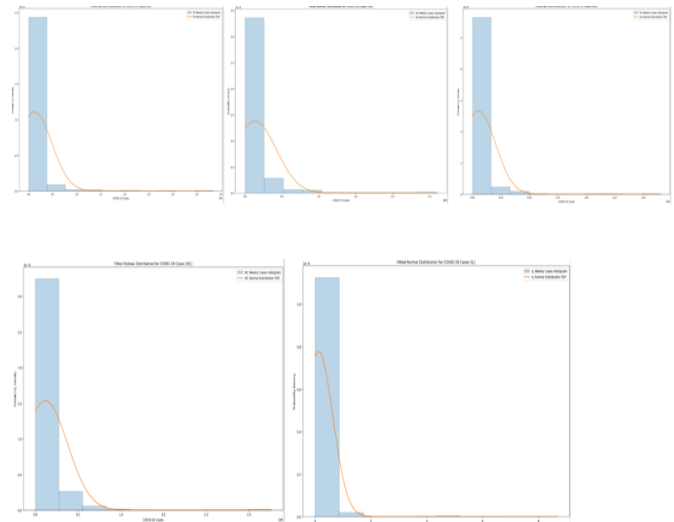


Fig. 38. Compararision with 5 other states

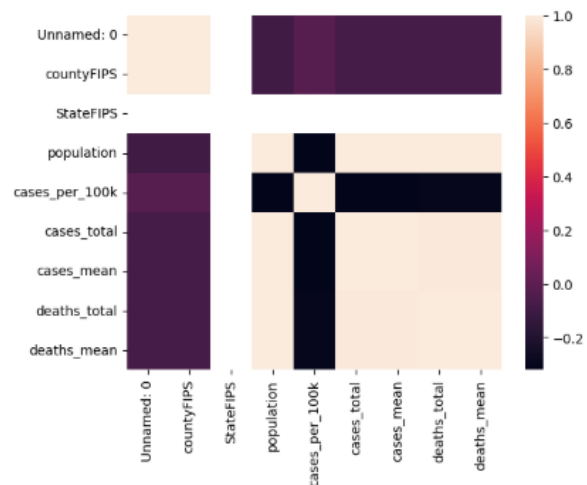


Fig. 39. Correlation Analysis Map

- Compare the Poisson statistics with those of the other five states.

**Correlation Analysis** Perform correlation analysis between enrichment variables (e.g., employment rates, median income, population density) and COVID-19 cases.

- Normalize the data for population.
- Compare correlations within a state (e.g., county-based data) and among different states.
- Identify any significant patterns or relationships.

#### A. Hypothesis Formulation and Testing

##### Hypothesis Questions:

- Do areas with higher average household income experience lower COVID-19 infection rates?
- Is there a significant difference in COVID-19 recovery rates between urban and rural counties?

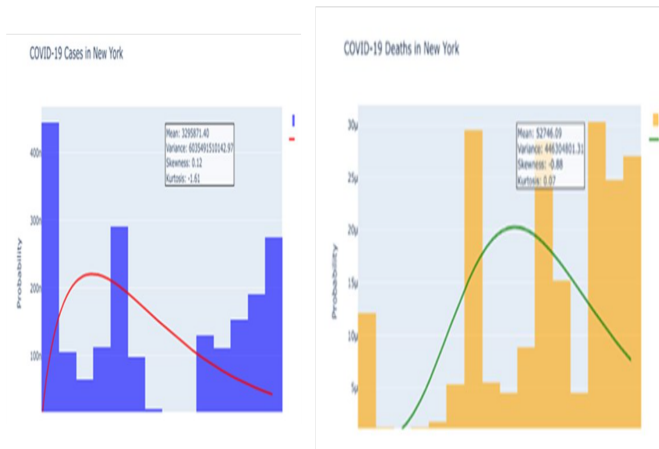


Fig. 40. Description of Graphs

- Does the level of educational attainment in a county affect the COVID-19 vaccination rates?

Three variables were selected for comparison, and statistical tests were conducted to validate or refute these hypotheses.

**result** Present the findings from your analyses, including graphs, tables, and statistical summaries. Discuss any observed trends, correlations, and differences between states.

### Employment Dataset

It is divided into three major parts. **Data set :**

- 1.Super\_covid19\_dataframe.csv.
2. County-ests-employment-and-wages.

#### Point 1 :

- Select the state and filter the State which is “NY”.
  - Plot the Histogram and negative binomial distribution for the state of NY.
  - Describing the Binomial distribution and finding the center, variance, skewness, and kurtosis of the Selected state.
  - Cases - Mean: 3295871.404743083.
  - Cases - Variance: 6035491510142.967.
  - Cases - Skewness: 0.11659171163707364.
  - Cases - Kurtosis: -1.6052417183660628.
  - Deaths - Mean: 52746.0885375494.
  - Deaths - Variance: 446304801.3123193.
  - Deaths - Skewness: -0.8833138420946183
  - Deaths - Kurtosis: 0.06607041860391405
- Description of graphs:-**
- The above graphs describe the Cases and Deaths in the NY state.
  - Cases graph is represented as a negative Binomial distribution because here variance exceeds the mean we can observe the same thing from the above data so we conclude it is a negative binomial distribution.
  - It is used to model the number of successes.

- In the question they mentioned only cases but I have done for both cases and deaths.
- The above-mentioned values are Mean, Variance, Skewness, and Kurtosis.
- Mean is the average number of cases per day during the given period.
- Variance is named as the Spread of the data set according to the above data there are many days with low cases and some days with very number of cases.
- About data tells that the head is in the left part and the tail is located at the right part which is known as right skew.
- Kurtosis denotes how fast the tail moved, meaning we can observe extreme peak values which are not fit in a normal distribution.

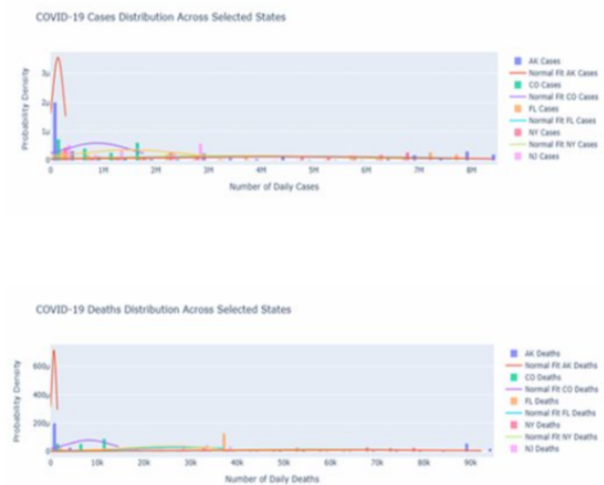


Fig. 41. Comparison of NY and Other Five(5) Countries

### Comparing NY state with 5 other states :-

- We can observe that sudden rise in peaks of every state at different periods in NY.
- Unexpected sudden rise in spike, overall may be stable but different in spike rise with different states.
- This also fits for Normal distribution where overdispersion data.
- Both deaths and cases got decreased gradually.

#### Point 2: A) Poisson Modeling:-

- $P(X=k)$  is the probability of observing exactly  $k$  events in the given interval.
- $k$  is the actual number of events (non-negative integer).
- $e$  is Euler's number (approximately 2.71828).
- $\lambda$  is the rate parameter, as mentioned above.

#### State: NY

- Mean Cases per 100k: 13898.350639640395
- Mean Cases per 100k: 16703.577786213777
- Mean Deaths per 100k: 149.78392263487544
- Total Cases: 4866754887,
- Total Deaths: 35767927

## State: CO

- Mean Cases per 100k: 14554.707835605719
- Mean Deaths per 100k: 170.07164469290598
- Total Cases: 1111856737,
- Total Deaths: 10391368

## State: NJ

- Mean Cases per 100k: 16332.407782125287
- Mean Deaths per 100k: 269.67656747182536
- Total Cases: 1894372763.
- Total Deaths: 32184948.

## State: AK

- Mean Cases per 100k: 15804.483685205905.
- Mean Deaths per 100k: 71.03823762752327.
- Total Cases: 178382300,
- Total Deaths: 904542 0.0.4 Description of Poisson distribution.

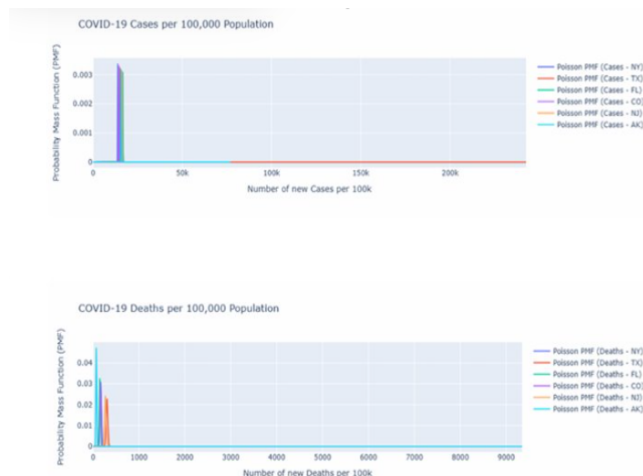


Fig. 42. Cases and Deaths per 1,00,000 population

- This distribution is used for events that occur independently of each other.
- It is denoted by.
- The graph suddenly shifts the highest peaks which results in more number of Cases registered and deaths happening.
- The Poisson distribution tells us more cases and deaths happen in the starting part later they got reduced.
- When we compare these Poisson graphs to normal distribution graphs both explain that there is a greater number of cases and deaths in the starting period and then they gradually increase and there is a sudden drop in the number of new cases and deaths as the time goes on they become stable.
- Highest Peak was encountered by Alaska, Florida, Colorado New Jersey, NewYorkand, and Texas for death due to climatic conditions deaths happened more in those states.

- Highest Peak was encountered by NewYork, Colorado, Texas, Alaska, NewJersey, and Florida the cases occurred in this pattern number of cases registered more due to the greater number of people going to work.

## Point 3 Correlation between the variables and plotting the scatter plot for the variables.

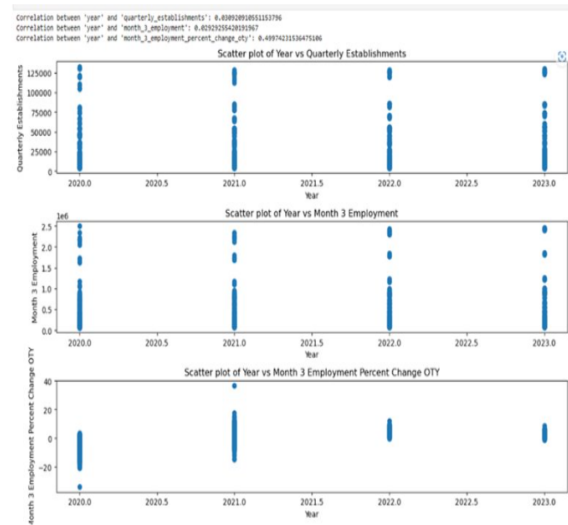


Fig. 43. Co-relation between the Variables and Scattering Points

## B) Hypothesis:

- Does Higher employment effects number of covid cases?
- Does a higher average weekly wage lead to higher COVID-19 case numbers?
- Do regions with more establishments see a more rapid increase in COVID-19 case numbers?

## Presidential Election Results

### 1. Fitting a Distribution to COVID-19 Case Data:

#### Data Overview:

The analysis started by extracting COVID-19 case data for the state of North Carolina from a larger dataset. Data was filtered from June 1, 2020, to January 3, 2021. A total of 217 days of data was analyzed, representing daily COVID-19 cases across multiple counties within the state.

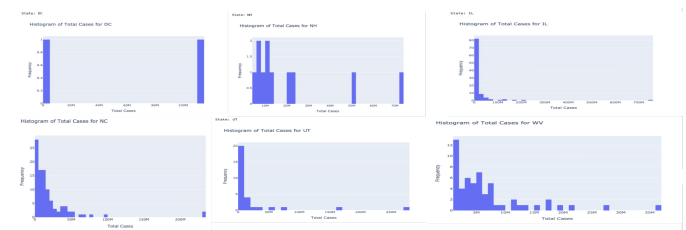


Fig. 44. Comparison of North Carolina with Other States

## Comparison with Other States:

- West Virginia: Although right-skewed, the distribution is fairly balanced, with a mix of low to moderately high case counts..
- Utah: Shows a right-skewed pattern with some outliers, indicating localized outbreaks and differences in reporting.
- Washington D.C.: Has little variation, likely due to consistent reporting or a concentration of cases in specific areas.
- New Hampshire: Displays a balanced distribution, suggesting effective control measures and steady case numbers across regions.
- Illinois: Mostly low case counts, but a few areas with high numbers, likely due to population density or localized surges.
- North Carolina: Positive skewness indicates most cases cluster at lower levels, with a few regions reporting significantly higher totals due to localized clusters. This suggests disparities in case spread, highlighting the need for targeted interventions in affected areas.

## 2. Modeling Poisson Distribution for COVID-19 Cases and Deaths:

Poisson Distribution for Case Counts:

Poisson distributions were modeled for daily new COVID-19 cases per 100,000 people in North Carolina and compared to five other states. Illinois, New Hampshire, Utah, West Virginia, and North Carolina have case distributions concentrated around lower numbers, reflecting lower averages compared to D.C. In contrast, D.C. stands out with a much higher mean of 78,134.64, shifting its distribution to the right and indicating a higher expected case count. The low peaks, typical of a Poisson distribution, suggest that while high case numbers can occur, they're unlikely to happen often.

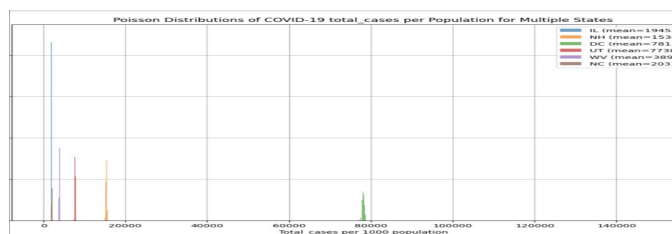


Fig. 45. Poisson Distribution of North Carolina with Other States

Poisson Distribution for Deaths:

A similar Poisson distribution was modeled for COVID-19 deaths, showing that states like Illinois, Utah, and North Carolina have low mean death counts, with most deaths concentrated around smaller numbers. In contrast, D.C. stands out with a much higher mean of 925.73, indicating a significantly higher number of expected deaths. The graph peaks reflect these patterns, with D.C.'s noticeably higher than the others.

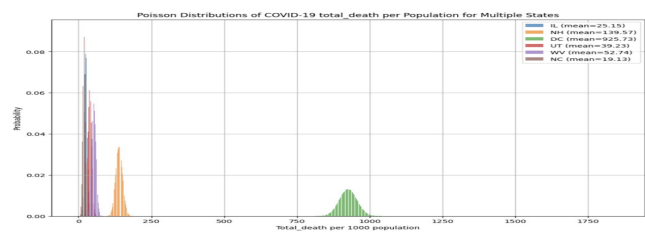


Fig. 46. Poisson Distribution of North Carolina with Other States

## Hypotheses:

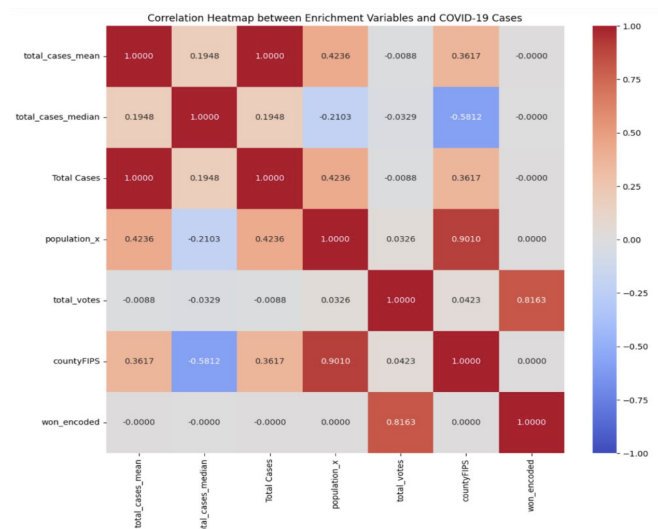


Fig. 47. Hypothesis Heatmap

The hypothesis we can derive from the correlation heatmap:  
Hypothesis 1: winning (won-encoded) and COVID-19 cases  
Null Hypothesis (H0): There's no significant relationship between election outcomes and the number of COVID-19 cases.

Alternative Hypothesis (H1): There is a significant relationship between election outcomes and COVID-19 cases, positive or negative

Hypothesis 2: total votes and average COVID-19 cases  
Null Hypothesis (H0): There's no significant relationship between the total number of votes and COVID-19 case levels.

Alternative Hypothesis (H1): There is a relationship between vote totals and COVID-19 case levels, either positive or negative.

Hypothesis 3: population size and COVID-19 cases  
Null Hypothesis (H0): There's no significant relationship between population size and the number of COVID-19 cases.

Alternative Hypothesis (H1): There is a significant positive relationship between population size and the number of COVID-19 cases.

## IV. STAGE IV - BASIC MACHINE LEARNING

Group task:

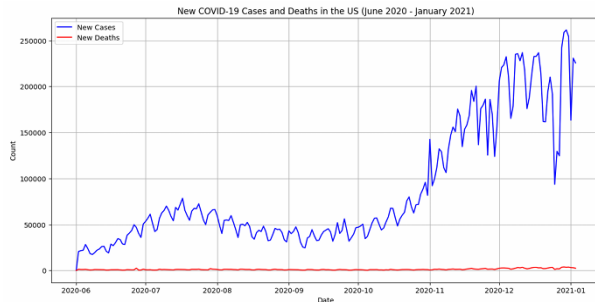
Step1:

Develop Linear regression models (polynomial with degree n) for predicting cases and deaths in the US.

a) Reading the super covid data.

b) Filtering the cases and deaths separately from the given dates 2020-6-1 to 2021-1-3

c) From the above data plot the graph with plotly for New cases and New deaths.



This plot shows the daily counts of new COVID-19 cases and deaths in the United States between June 2020 and January 2021.

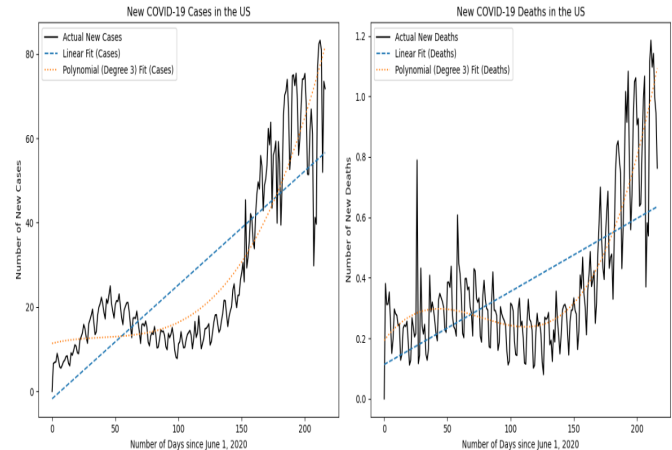
- **Blue Line (New Cases):** The number of new cases per day. There is a clear upward trend, particularly from October 2020 onward, where cases rise sharply and fluctuate significantly around higher daily counts, peaking near 250,000 cases per day in late December 2020.
- **Red Line (New Deaths):** The number of new deaths per day. This line remains relatively low and stable compared to the cases line, with small fluctuations and peaks, but it does not increase as dramatically as the cases.

Fig. 48. New cases and deaths of US:

New cases and deaths in the US:

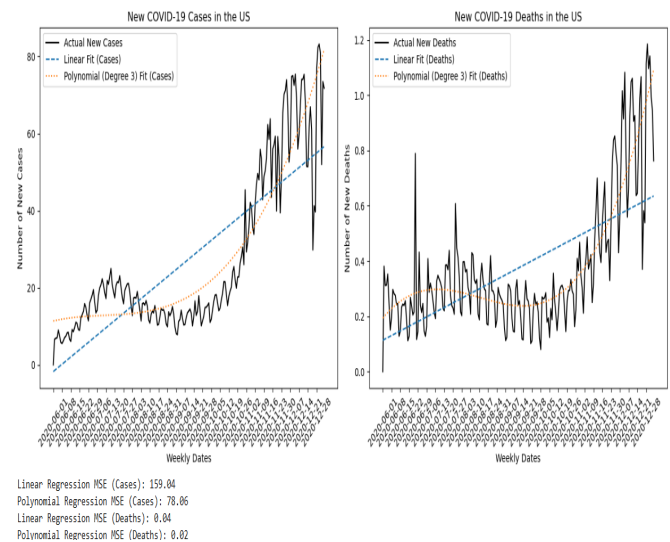
- The plot highlights the significant surge in cases during late 2020, while deaths show a much smaller increase relative to cases, suggesting either mitigation efforts that reduced death rates or delays in the rise of deaths compared to cases.
- Discuss bias versus variance trade-off. (10 points) The plot trend line for the data along with the forecast of 1 week ahead (i.e. predict 2021.01.04 - 2021.01.10). (10 points).
- The code includes two functions that analyze and predict COVID-19 data using regression models. The first function applies linear and polynomial regression to historical case and death data and then compares the models' accuracy using error metrics.
- It displays both actual and predicted values in separate plots. The second function builds on this by forecasting future cases and deaths based on the models.
- Both functions generate detailed plots and output the predicted values for the specified future dates.
- USA plotting for cases and deaths for both Linear and polynomial model and RMSE also forecasting for cases and death Jan 4, 2021 - Jan 10, 2021, both Linear and Polynomial (degree 3)

Step 2: Calculate and report Root Mean Square Error (RMSE) for days.



Linear Regression MSE (Cases): 159.04  
Polynomial Regression MSE (Cases): 78.06  
Linear Regression MSE (Deaths): 0.04  
Polynomial Regression MSE (Deaths): 0.02

Fig. 49. Calculate and report Root Mean Square Error (RMSE) for days



Linear Regression MSE (Cases): 159.04  
Polynomial Regression MSE (Cases): 78.06  
Linear Regression MSE (Deaths): 0.04  
Polynomial Regression MSE (Deaths): 0.02

Fig. 50. Calculate and report Root Mean Square Error (RMSE)

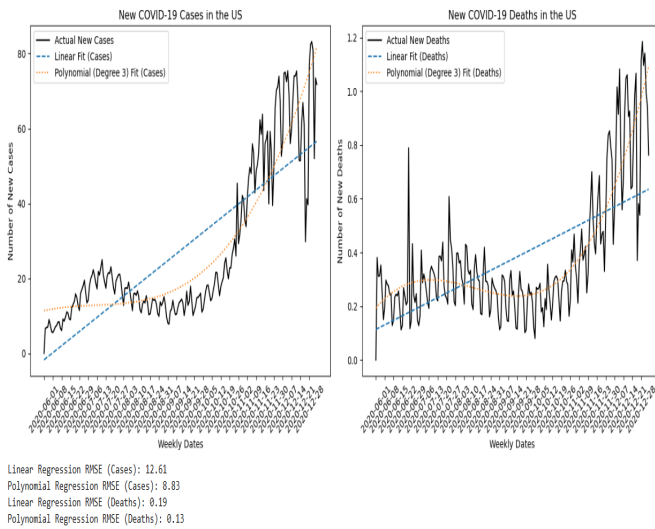


Fig. 51. Calculate and report linear regression (MSE)

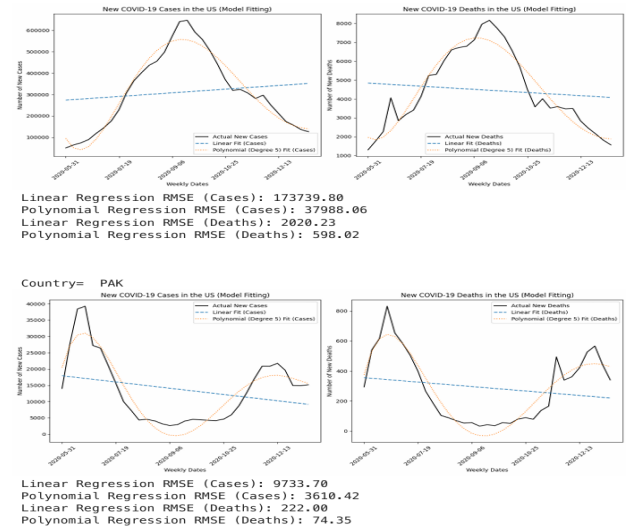


Fig. 53. IND,PAK

### Step 3:

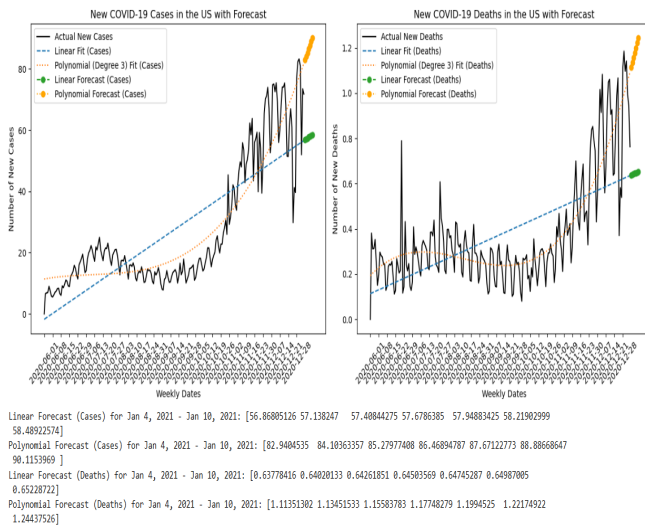


Fig. 52. Plot trend line along for the data along with the forecast of 1 week ahead (i.e. predict 2021.01.04 - 2021.01.10)

The plot trend line along for the data along with the forecast of 1 week ahead (i.e. predict 2021.01.04 - 2021.01.10)

### Step 4:

Describing the trends as compared to other countries considering "IND", "PAK", "AFG", "GRL", "CHN", and "AUS" countries.

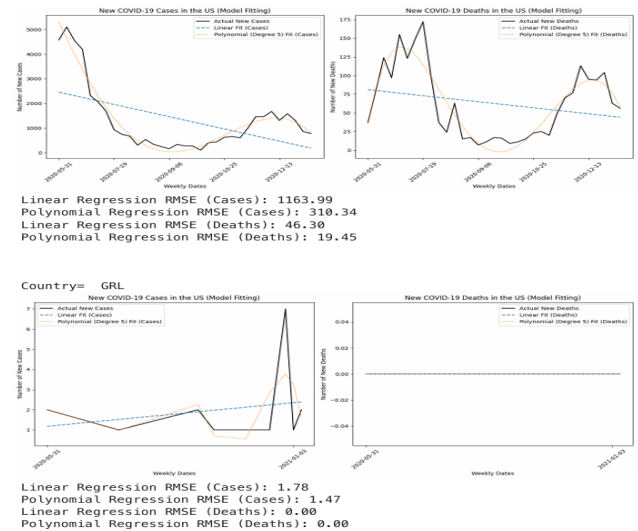


Fig. 54. AFG , GRL

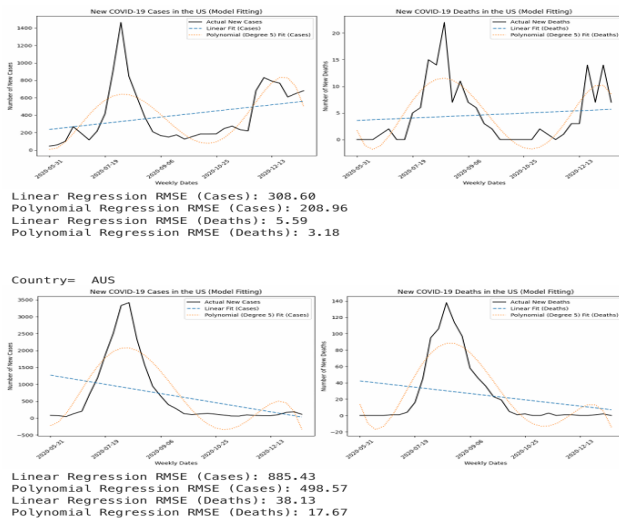


Fig. 55. CHN, AUS

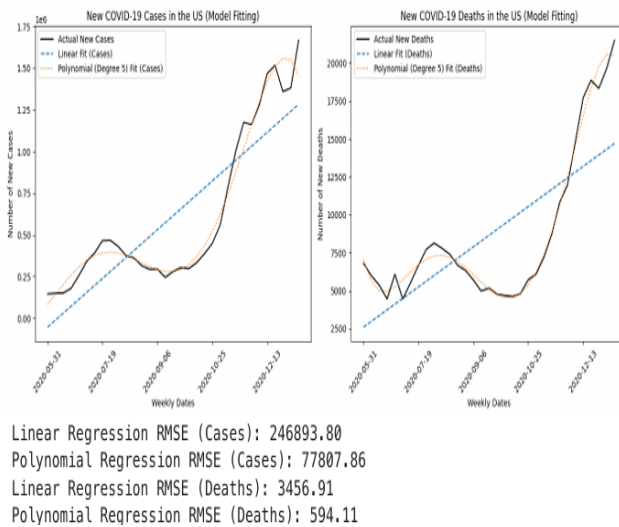


Fig. 56. USA

### 1.COVID-19 Cases Trends:

- PAK & AFG: Both countries show a peak followed by a decline and some resurgence. The polynomial model better captures the fluctuations than the linear model.
- GRL: With minimal fluctuation and low data, both models fit well.
- CHN & AUS: The polynomial model provides a better fit, capturing fluctuations in cases more accurately.
- USA: High variation in cases; the polynomial model fits much better, capturing the larger fluctuations.

### 2. COVID-19 Deaths Trends:

- PAK & AFG: Both show an initial peak followed by a decline and some increases. The polynomial model tracks these shifts more closely than the linear model.
- GRL: Virtually no deaths; both models fit perfectly with minimal error.

- CHN & AUS: Polynomial models capture the rise and fall of deaths more accurately.
- USA: Polynomial models provide a better fit, closely matching the actual death data.

### 3. Model Performance (RMSE values).

- GRL: Lowest RMSE due to minimal data fluctuations.
- PAK & AFG: Higher RMSE values, with Pakistan showing the highest errors due to more data variability.
- CHN & AUS: Polynomial regression gives a better fit with lower RMSE values for both cases and deaths.
- USA: The highest RMSE values, especially for cases, reflect the larger and more complex dataset.
- Polynomial regression generally provides a better fit, especially in capturing fluctuations in COVID-19 trends.
- Linear regression oversimplifies the data and does not capture the variability in countries with large fluctuations.
- RMSE values are higher in countries with more data, like the USA, and lower in countries with less variation, like Greenland.

## Census Demographic ACS

In this task, we need to utilize Linear and Non-Linear (polynomial with degree 3) regression models to compare trends for the state of Arizona. After that, we need to calculate the error using RMSE. We observe that the RMSE value is relatively large, which is likely due to the use of the original dataset rather than a normalized one. By normalizing the dataset, we can expect to achieve smaller RMSE values compared to the current results. After identifying which counties are most at risk I observed that Maricopa and Pima Counties show a consistent increase. Yuma County exhibits significant growth initially but stabilizes later. Navajo and Mohave Counties show similar growth.

Below is my hypothesis test:

### Hypothesis 1:

Null Hypothesis (H0): There is no significant correlation between the 5 to 9-year-old age group population and the number of COVID-19 cases.

Alternative Hypothesis (H1): There is a significant correlation between the 5 to 9-year-old age group population and the number of COVID-19 cases.

Variables: 5 to 9 years age group population and cases (from US COVID dataset)

### Hypothesis 2:

Null Hypothesis (H0): There is no significant correlation between the female population and the number of COVID-19 cases.

Alternative Hypothesis (H1): There is a significant correlation between the female population and the number of COVID-19 cases.

Variables: Female population (from US COVID dataset)

### Hypothesis 3:

Null Hypothesis (H0): There is no significant correlation between the age groups under 5 years population and the

number of COVID-19 cases.

Alternative Hypothesis (H1): There is a significant correlation between the age groups under 5 years population and the number of COVID-19 cases.

Variables: Under 5 years age group population and cases (from US COVID dataset)

## ACS Social, Economic, and Housing

### INTRODUCTION

This section explains the motivation and importance of analyzing COVID-19 cases in relation to socio-economic factors. It highlights the objective of the study which includes the identification of top counties affected by COVID-19, and the prediction of cases using regression models.

### V. CONVERT DATE COLUMNS

Transform columns to datetime format for cases and deaths data.

### VI. FILTER DATE RANGE

Focus on data from June 1, 2020, to January 3, 2021.

### VII. CONCATENATE IDENTIFIER COLUMNS

Re-attach identifier columns to the filtered cases and deaths data.

### IDENTIFY TOP COUNTIES

#### A. Identify Top 5 Counties

Determine the counties with the highest number of cases for further analysis.

### REGRESSION MODELS

#### B. Date Transformation

Convert dates into numerical format (days since June 1, 2020) for modeling.

#### C. Linear Regression

Fit a linear model to predict new cases and deaths based on days since June 1, 2020.

#### D. Polynomial Regression

Fit a polynomial (degree 3) model to capture more complex trends.

## PREDICTIONS AND VISUALIZATION

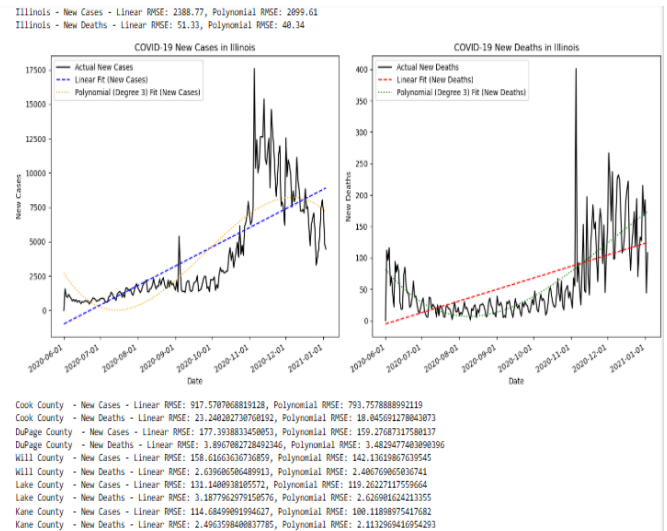


Fig. 57. New COVID-19 cases and deaths in IL

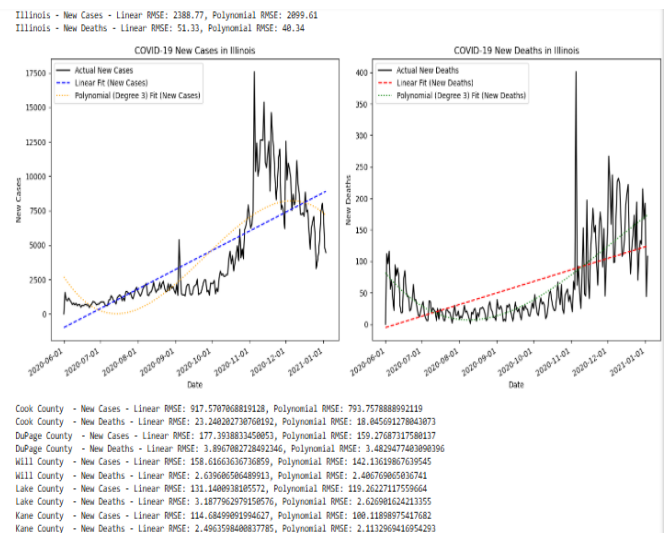


Fig. 58. New COVID-19 cases and deaths in NY state

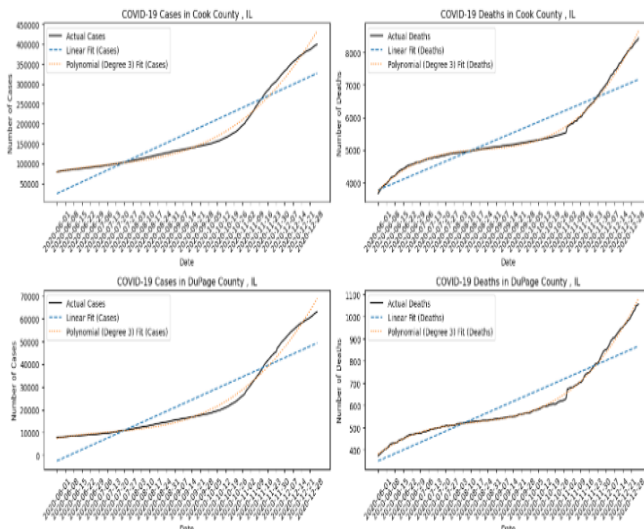


Fig. 59. New COVID-19 cases in top 5 counties RMSE

### E. Generate Predictions

Use both models to predict daily cases and deaths.

### F. Plot Results

Compare actual data to model predictions with separate graphs for cases and deaths for each of the top counties.

## MODEL EVALUATION

### G. RMSE Calculation

Calculate RMSE for each model to evaluate prediction accuracy:

- Linear RMSE: Measures error for the linear model.
- Polynomial RMSE: Measures error for the polynomial model.

## HYPOTHESIS TESTING

We examine three hypotheses using statistical tests:

### Hypothesis 1: Employment Rates and COVID-19 Case Increase

**Hypothesis:** Counties with higher unemployment rates experienced a greater increase in COVID-19 cases.

**Method:** Conduct a two-sample t-test between the employment rate estimates and COVID-19 cases.

**Interpretation:** If the p-value is below 0.05, we may conclude a significant relationship between unemployment rates and COVID-19 case increases.

### Hypothesis 2: Employment Estimate Margin of Error and COVID-19 Deaths

**Hypothesis:** Regions with higher margins of error in employment estimates have less reliable COVID-19 data trends.

**Method:** Use a two-sample t-test between the employment margin of error and COVID-19 deaths.

**Interpretation:** A low p-value indicates that variations in employment data reliability could be associated with inconsistencies in COVID-19 death rates.

### Hypothesis 3: Poverty Rates and COVID-19 Case Reporting

**Hypothesis:** Counties with higher poverty levels reported fewer COVID-19 cases due to lower testing rates.

**Method:** A two-sample t-test between average wage estimates and COVID-19 cases.

**Interpretation:** A significant p-value suggests that lower income levels may correlate with fewer reported cases, implying disparities in testing or reporting.

## Employment Dataset

Step 1: X-Axis - number of days, Y-Axis number of new cases and deaths. Calculate error using RMSE.

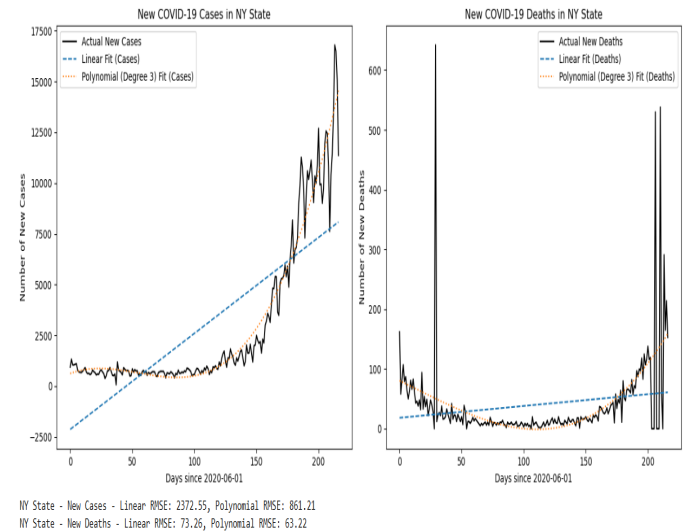


Fig. 60. New covid cases and deaths in NY state

Step 2: Identify which counties are most at risk. Model for top 5 counties with cases within a state and describe their trends.

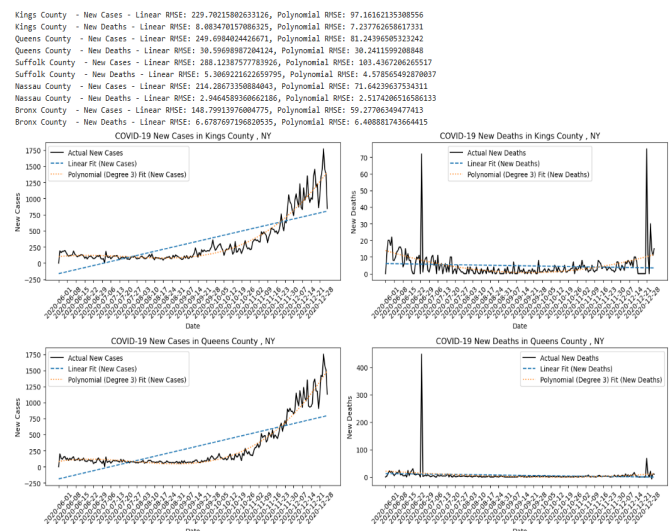


Fig. 61.

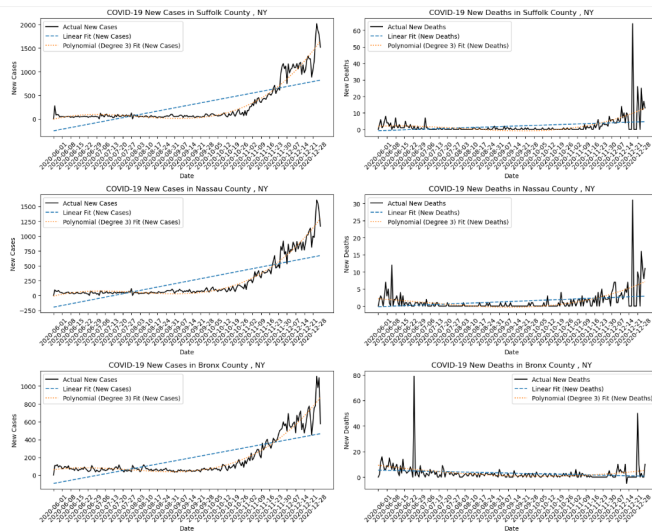


Fig. 62. New COVID cases and deaths of counties in NY state

Step 3: Identify which counties are most at risk. Model for top 5 counties with cases within a state and describe their trends

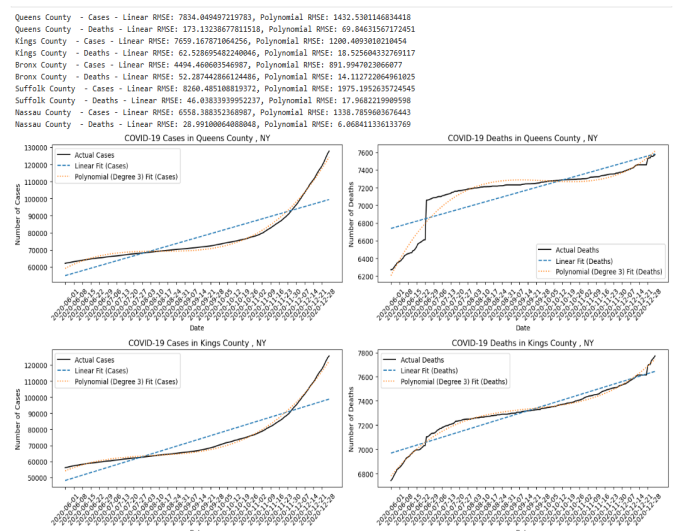


Fig. 63. Top most risk of counties in NY state

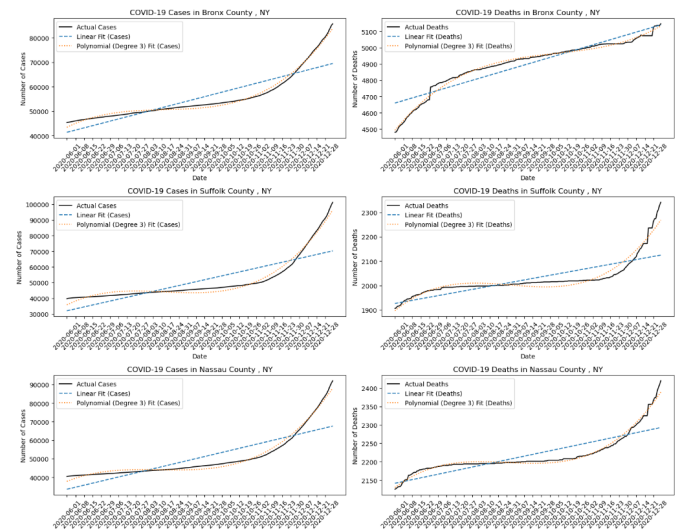


Fig. 64. Top most risk of counties in NY state

Step 4: 7 day Trend line prediction

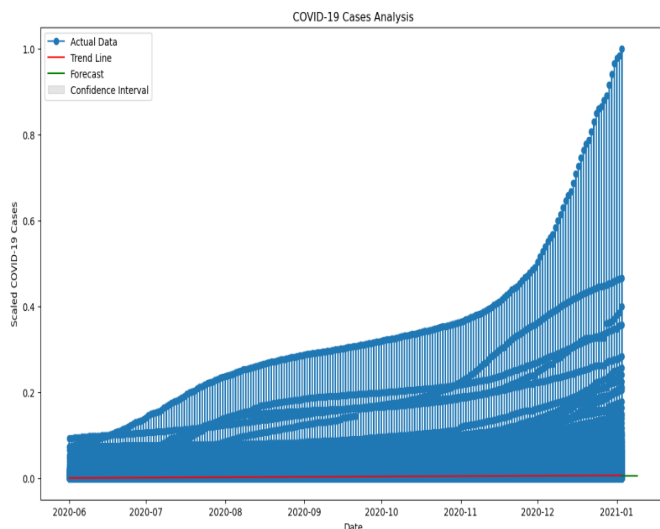


Fig. 65. 7-day Trend line prediction

### Presidential Election Results

In this task, we need to utilize linear and non-linear regression models to compare trends in North Carolina. After that, we need to calculate the error using RMSE. Below the graph displays new COVID-19 cases and deaths in North Carolina from June 2020 to January 2021. The green line shows a sharp rise in cases starting around September, peaking in December. The blue line, representing new deaths, follows a similar pattern but with a less dramatic increase compared to the surge in cases. Overall, the data reveals a rapid spread of COVID-19 and a noticeable rise in deaths towards the end of 2020.

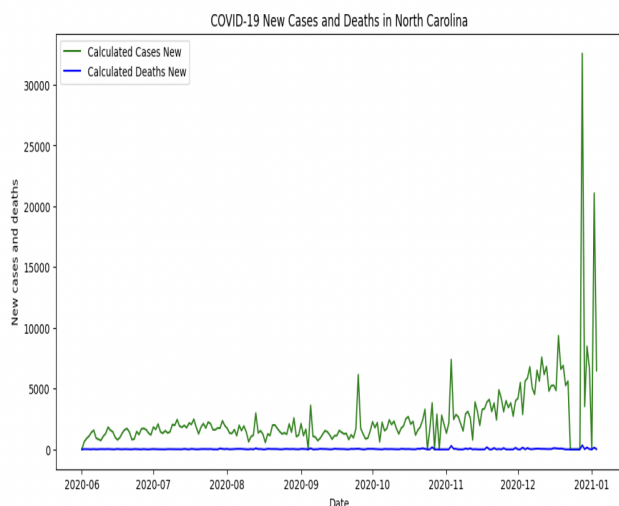


Fig. 66. new COVID-19 cases and deaths in North Carolina from June 2020 to January 2021

The graphs show fluctuations in new COVID-19 cases and deaths in North Carolina, with both following an overall upward trend. The linear regression lines reflect this general

increase, while a degree 3 polynomial regression was used to capture the data's complexity without overfitting, offering a good balance of accuracy and simplicity.

RMSE for Linear Regression (New Cases), NC: 2637.93

RMSE for Linear Regression (New Deaths), NC: 40.22

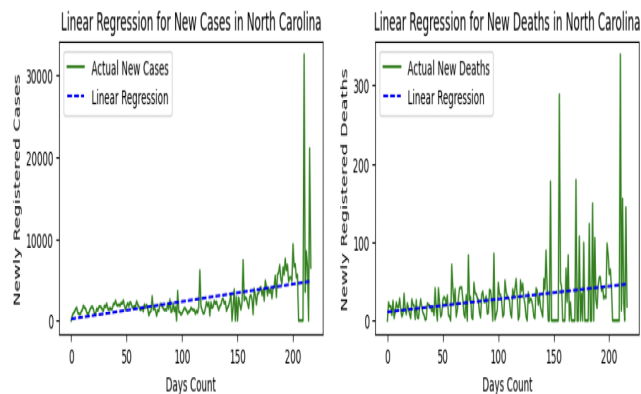


Fig. 67. trends of new COVID-19 cases and deaths in North Carolina - Linear Regression Model

The graphs show a sharp rise in COVID-19 cases and deaths in North Carolina. The polynomial regression models (red dashed lines) provide a more accurate fit than linear regression, effectively capturing the trends and fluctuations, with lower RMSE values, particularly for deaths

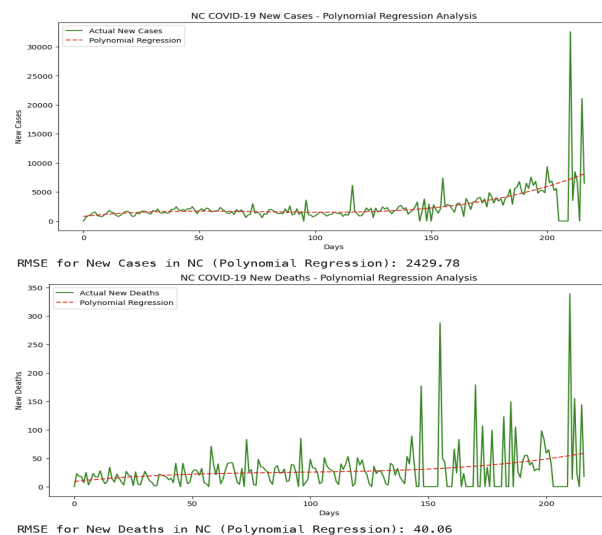


Fig. 68. two graphs showing the trends of new COVID-19 cases and deaths in North Carolina - Polynomial Regression Model

We also applied linear and polynomial regression to the top 5 counties in North Carolina to Analyse the Trend. Below is my hypothesis test:

### Hypothesis 1:

Null Hypothesis (H0): There is no significant correlation between the political party of the winning candidate in a

state and the total number of COVID-19 cases reported in that state.

Alternative Hypothesis (H1): There is a significant correlation between the political party of the winning candidate in a state and the total number of COVID-19 cases reported in that state.

Analysis: Correlation coefficient: 0.068

P-value: 0.16

Accept the null hypothesis: There is no significant correlation between population and total cases.

### Hypothesis 2:

Null Hypothesis (H0): There is no significant relationship between voter turnout in a state and the total number of COVID-19 cases reported in that state.

Alternative Hypothesis (H1): There is a significant relationship between voter turnout in a state and the total number of COVID-19 cases reported in that state.

Analysis: Fail to reject the null hypothesis: There is no significant difference in COVID-19 case rates between Republican and Democrat counties.

### Hypothesis 3:

Null Hypothesis (H0): There is no significant correlation between the population size of a state and the total number of COVID-19 cases reported in that state.

Alternative Hypothesis (H1): There is a significant correlation between the population size of a state and the total number of COVID-19 cases reported in that state.

Analysis: Correlation coefficient: -0.027

P-value: 0.5855931118162806 Accept the null hypothesis:

There is no significant relationship between voter turnout and COVID-19 case counts.

## VIII. - DASHBOARD

Task 1:

- Allow for selection of date to show the trend of COVID-19 cases and deaths.
- Allow for linear or log mode selection on the number of cases and deaths.
- Incorporate your best model prediction trend line - Linear/Non-Linear.

Task 2:

- Trend - Plot the trend line using the moving average. Use 7-day moving average.
- Allow for selection of multiple states on the same graph.

**Introduction** The *COVID-19 Data Analysis and Interactive Dashboard* project aims to provide an intuitive platform for visualizing COVID-19 cases and deaths trends across the United States. The final stage of the project involves creating an interactive dashboard using Plotly and Dash, allowing users to explore data interactively with various filtering options. The dashboard integrates features for analyzing trends, calculating moving averages, and visualizing predictions, making it a valuable tool for understanding the pandemic's progression.

**Goals** The main objectives of the project are:

- To create a dashboard that visualizes COVID-19 cases and deaths.
- To enable users to interact with the data by selecting date ranges, states, and performance metrics.
- To integrate features for displaying trendlines, moving averages, and linear or logarithmic scales.
- To predict future trends using simple regression techniques.

### Technologies Used

#### A. Python Libraries

- **Pandas:** For data manipulation and preprocessing.
- **NumPy:** For numerical operations, including polynomial regression.
- **Plotly:** For creating visually appealing and interactive plots.
- **Dash:** For building the interactive web application.

### Dash Components

- `dcc.DatePickerSingle`: For selecting date ranges.
- `dcc.Checklist`: For choosing performance metrics.
- `dcc.Dropdown`: For selecting states.
- `dcc.Graph`: For rendering dynamic plots.

### Data Sources

#### B. COVID-19 Cases Data

- **File:** `covid_confirmed_usafacts.csv`
- Contains daily cumulative cases data for each U.S. County.

### COVID-19 Deaths Data

- **File:** `covid_deaths_usafacts.csv`
- Contains daily cumulative deaths data for each U.S. County.

### Steps and Implementation

#### C. Data Preprocessing

The raw data was preprocessed for better usability:

- **Remove Unnecessary Columns:** Dropped columns like `countyFIPS`, `County Name`, and `StateFIPS`.
- **Set State as Index:** Reformatted the data to use the `State` column as the index.
- **Transpose the Data:** Transposed the dataset to switch rows and columns, making dates the index.
- **Group by State:** Aggregated data for each state by summing values.
- **Rename Columns:** Simplified column names for consistency.

#### D. Time Series Preparation

- Converted indices of the cases and deaths datasets to `datetime` format for easier time-based filtering.
- Designed components like `DatePickerSingle`, `Checklist`, and `Dropdown` for dynamic user interactions.

## E. Dashboard Layout

The dashboard was structured into the following components:

- Title: "COVID-19 Dashboard".
- Date Pickers: For selecting start and end dates.
- Mode Selector: For toggling between linear and logarithmic scales.
- Performance Checklists: For selecting options like actual values, trendlines, and moving averages.
- State Selector Dropdowns: For choosing states to visualize.
- Graphs: Separate plots for cases and deaths.

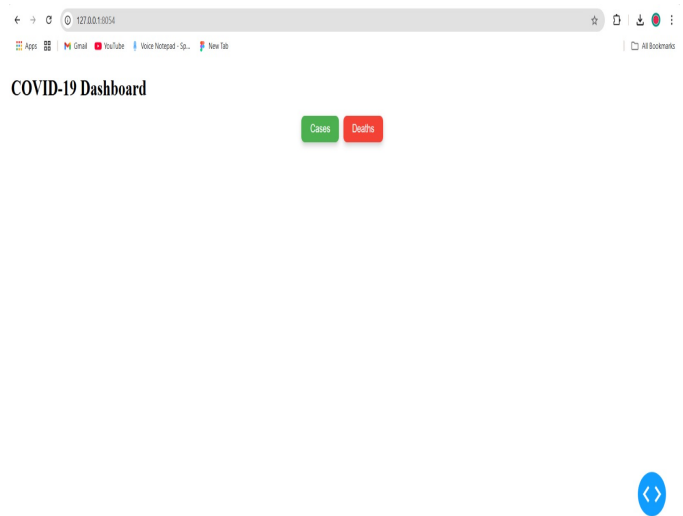


Fig. 70. Dashboard

## F. Interactive Callbacks

Dash callbacks were used to update the graphs dynamically based on user inputs:

- **Cases Graph Callback:** Filters and visualizes cases data based on selected date range, states, and performance options.
- **Deaths Graph Callback:** Filters and visualizes deaths data with similar functionalities.

## G. Trendline and Prediction

Using polynomial regression:

- **Trendline:** Generated for each state's data.
- **Prediction:** Forecasted the next 7 days based on the trendline.

## H. Running the Dashboard

The app is launched on a specified port, making it accessible via a web browser.

### Features

- **Interactive Date Range Filtering:** Users can explore data within custom date ranges.
- **State-Wise Data Selection:** Multiple states can be compared simultaneously.
- **Performance Metrics:** Options for actual values, trendlines, and moving averages.
- **Logarithmic Scale Option:** Visualize data in linear or logarithmic scales.
- **Trendline Predictions:** Forecast upcoming trends for selected states.

### Dashboard Screens

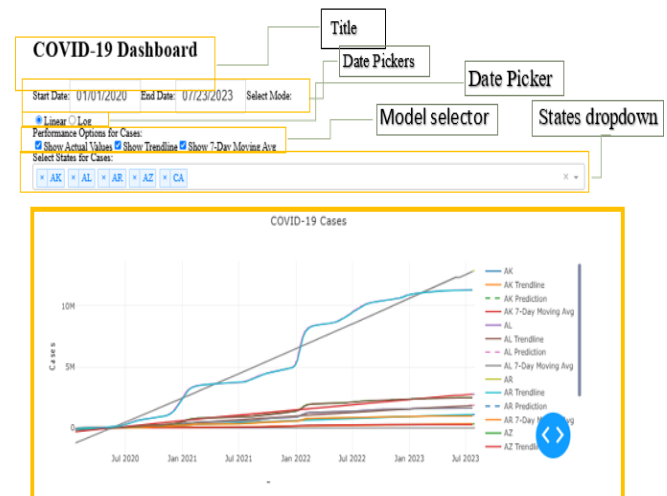


Fig. 69. Over View of Dashboard

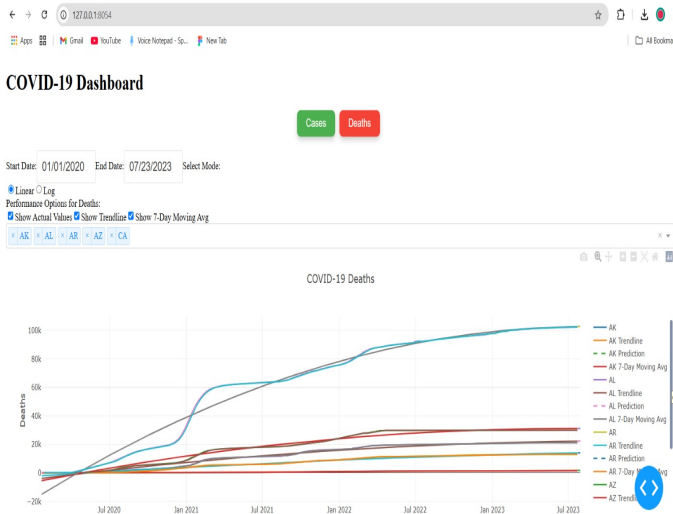


Fig. 71. Final Dashboard

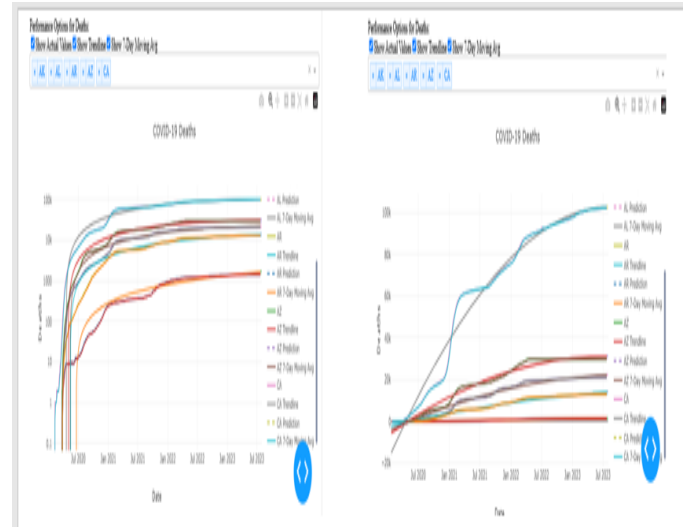


Fig. 73. Linear graph VS Log graph

### I. Screen 1: Initial Dashboard View

- **Description:** The dashboard opens with default selections.
- **Default Settings:**
  - Start Date: 2020-01-01.
  - End Date: 2023-07-23.
  - Mode: Linear.
  - Performance Options: All selected (actual values, trendline, and 7-day moving average).
  - States: Displays the first five states by default for both cases and deaths.
- **Graph View:** The cases graph and deaths graph display trends based on the full date range and default settings.

### J. Screen 2: Customized Date Range

- **Description:** Users can customize the date range to focus on specific periods.
- **User Interaction:**
  - Date pickers are adjusted to a shorter time frame, e.g., 2021-06-01 to 2021-12-31.
- **Graph View:** The graphs update dynamically to reflect the selected date range. Peaks and trends for the specified period are highlighted.

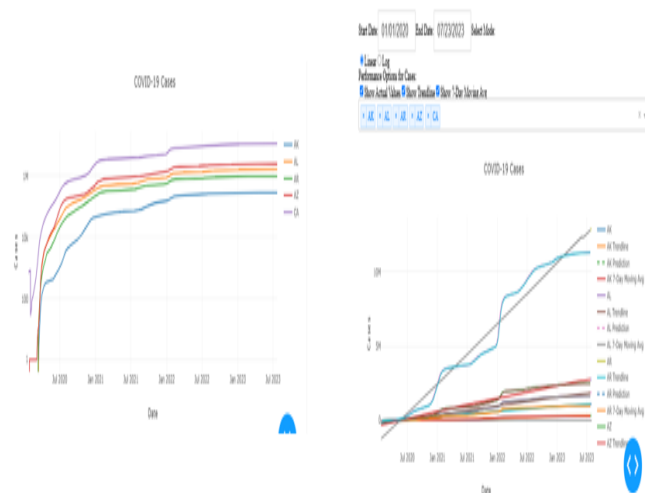


Fig. 72. Graphs

### K. Screen 3: Logarithmic Mode

- **Description:** Demonstrates the logarithmic view for better visualization of data spanning several magnitudes.
- **User Interaction:**
  - The mode selector is switched to Log.
- **Graph View:** Both graphs now display data using a logarithmic scale, making it easier to observe trends for smaller values.

*L. Screen 4: Customized Performance Options*

- **Description:** Users can modify the data visualization by selecting specific performance options.
- **User Interaction:**
  - Uncheck 7-Day Moving Average and Trendline to focus on raw data.
- **Graph View:** The graphs show only actual values, simplifying the view for detailed observation of data points.

*M. Screen 5: Multi-State Comparison*

- **Description:** Compare COVID-19 trends across multiple states.
- **User Interaction:**
  - Users select a different set of states (e.g., California, Texas, New York) in the dropdowns.
- **Graph View:** The graphs update to display data for the selected states, allowing comparisons across regions.

ACKNOWLEDGEMENT

We would like to sincerely thank Professor Qianqian Tong for her invaluable guidance throughout the development of this project. Her expertise and support have been instrumental in helping us understand the complexities of data analysis. Through her mentorship, we gained a deeper insight into various analytical methods and learned how to effectively interpret data. Her guidance has not only shaped our work on this project but also enriched our understanding of the field as a whole.